National Polytechnic School

Department of Electrical Engineering

Design of AC machines

Mini-Project Report

Design of a Three-Phase Squirrel-Cage Induction Machine : Development of a PyQt6-Based Calculation and 2D Design Guide with Complementary 3D Visualization

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Chapitre 1

Introduction

This project involves the development of two complementary software applications dedicated to the design and analysis of a three-phase squirrel cage induction machine. The objective is to provide a complete computational and graphical framework covering the **entire machine**, including the stator, rotor, air-gap, and overall geometric configuration.

The first application is a desktop-based computational tool developed using **PyQt6**. It focuses on the analytical and numerical calculation of the induction machine parameters, integrating electromagnetic modeling, geometric dimensioning, material selection, and performance evaluation into a unified and automated workflow.

The second application is a **web-based design tool developed using HTML**, dedicated to the **2D and 3D geometric design and visualization of the complete machine**. This application allows interactive representation of the stator, rotor, slots, air-gap, and housing, enabling users to visually analyze the influence of design parameters on the overall machine structure.

1.1 Project Scope

The project addresses the multidisciplinary nature of induction machine design, which requires balancing several interacting constraints :

- **Electrical performance** : Flux distribution, current density, losses, efficiency, and power factor
- **Geometric constraints** : Stator and rotor dimensions, slot shapes, air-gap length, and manufacturability
- **Thermal considerations** : Heat dissipation, cooling capability, and temperature rise
- **Mechanical aspects** : Structural integrity and dimensional consistency

1.2 Key Features

The developed applications provide :

- Automated calculation of stator and rotor dimensions using a PyQt6-based computational interface
- Electromagnetic and performance analysis of the complete induction machine
- Material selection from an extensive database
- Comprehensive constraint checking and design validation
- A user-friendly desktop application with detailed numerical reporting

— A separate HTML-based application for interactive **2D and 3D visualization of the entire machine**

1.3 Technical Approach

The adopted methodology combines :
— Analytical electromagnetic theory for machine modeling
— Numerical optimization for dimensioning and parameter selection
— Empirical design rules validated by industrial practice
— Modular software architecture separating computation and visualization
— Web technologies (HTML) for full-machine geometric modeling and graphical interaction

1.4 Significance

This project bridges the gap between theoretical machine design principles and practical engineering implementation by offering :
— **Time savings** : Reducing design cycles from days to hours
— **Accuracy improvement** : Minimizing manual calculation errors
— **Design coherence** : Ensuring consistency between electrical calculations and geometric representation
— **Educational value** : Serving as a learning platform for induction machine design through both numerical analysis and 2D/3D visualization

The following sections present the theoretical foundations, software architecture, and validation of both applications, demonstrating their capability to produce optimized, manufacturable, and high-performance squirrel cage induction machines.

Chapitre 2

Stator Design Algorithmic Process

2.1 Introduction

The stator is the stationary component of an induction motor, responsible for generating the rotating magnetic field that drives the rotor. Designing a stator involves a complex interplay between magnetic capabilities (flux), electrical capabilities (current), and mechanical dimensions.

The Python application documented here automates this process using a **Forward Design Approach**. This means the software starts with the desired output (e.g., Power, Voltage) and sequentially calculates the required physical dimensions to achieve that output, iterating only when necessary to satisfy integer constraints (such as the number of turns in a coil).

This report is written as a **translation of the program's algorithm** into engineering reasoning : each module states the physical objective, the governing equations, the constraints, and the decision logic used by the software.

2.2 Module 1 : Empirical Design Targets (Specific Loadings and Performance Estimates)

Before specific dimensions can be calculated, the algorithm must estimate the electromagnetic "stress" the materials can handle. These limits are derived from empirical data (historical data from successfully built machines).

2.2.1 Algorithm Overview (Program Steps)

1. Read the internal empirical datasets (abaques) for B_{av} and q versus P_{rated}.
2. Interpolate B_{av} and q at the user-specified rated power.
3. If not explicitly specified by the user, estimate η and $\cos\varphi$ from regression formulas.
4. Compute the synchronous speed n_s from supply frequency f and pole count P.

2.2.2 Symbols and Parameters

— P_{rated} : rated mechanical output power [kW]
— B_{av} : specific magnetic loading (average air-gap flux density) [T]
— q : specific electric loading (ampere-conductors per meter) [A/m]
— η : efficiency estimate [-]

— $\cos\varphi$: power factor estimate [-]
— f : supply frequency [Hz]
— $2p$ (or P) : number of poles ; $p = \frac{P}{2}$
— n_s : synchronous speed [RPM]
— K_w : winding factor [-]

2.2.3 Definitions

— **Specific Magnetic Loading (B_{av}) :** The average magnetic flux density over the air gap surface. A higher B_{av} means the machine is magnetically stronger but may saturate the iron core. Measured in Tesla [T].
— **Specific Electric Loading (q or ac) :** The amount of electrical current flowing along the inner surface of the stator per unit of length. A higher q increases torque but generates more heat. Measured in Ampere-conductors per meter [A/m].

2.2.4 Algorithmic Process

The software references internal lookup tables (Abaques) to select optimal values for B_{av} and q based on the rated power (P_{rated}).

1. **Interpolation :** The code linearly interpolates values from Table 7.1 to find the precise B_{av} and q for the specific input power.
2. **Performance Estimation :** The efficiency (η) and power factor ($\cos\phi$) are estimated using logarithmic regression formulas. These values predict how much electrical energy is converted to mechanical work versus heat.

2.2.5 Empirical Reference Tables

The following table represents the internal dataset (Abaques Table 7.1) used by the software for interpolation.

TABLE 2.1 – Specific Magnetic and Electric Loading vs. Power (Table 7.1)

Rated Power (kW)	**Magnetic Loading B_{av} (T)**	**Electric Loading q (A/m)**
1	0.35	16,000
2	0.38	19,000
5	0.42	23,000
10	0.46	25,000
20	0.48	26,000
50	0.50	29,000
100	0.51	31,000
500	0.53	33,000

2.2.6 Performance Estimation Formulas

If specific targets are not provided by the user, the software estimates performance using the following empirical regressions derived from standard motor characteristics :

$$p = \frac{\text{Poles}}{2}, \quad n_s = \frac{60f}{p} \quad \text{[RPM]} \tag{2.1}$$

Efficiency (η) :

$$\eta = 1 - \left[0.01 + 0.03 \cdot P_{rated}^{-0.03} + 0.1 \cdot \ln(p)\right] \tag{2.2}$$

Power Factor ($\cos\phi$) :

$$\cos\phi = 0.84 - 0.04 \cdot \ln(P_{rated}) + 0.02 \cdot \ln\left(\frac{n_s}{p}\right) \tag{2.3}$$

2.2.7 Winding Factor Calculation (K_w)

The winding factor K_w, a dimensionless parameter between 0 and 1, quantifies the effectiveness of the stator winding in producing a sinusoidal magnetomotive force (MMF). A higher K_w indicates more efficient use of copper and a stronger, cleaner fundamental magnetic field for a given current. In this application, K_w is no longer selected from a fixed table but is calculated dynamically based on the motor's geometric and winding parameters. This approach provides greater accuracy and flexibility, covering all potential winding designs, including fractional-slot configurations.

Theoretical Foundation

The total winding factor K_w is the product of two independent factors that account for the spatial distribution and the pitch (chording) of the coils :

$$K_w = k_d \times k_p \tag{2.4}$$

where :
— k_d is the **distribution (or breadth) factor**.
— k_p is the **pitch (or chording) factor**.

A third factor, the skew factor k_{skew}, may be included for motors with skewed slots or rotors ($K_w = k_d \times k_p \times k_{skew}$). For simplicity and as a common default, this application assumes no skew ($k_{skew} = 1$).

Distribution Factor, k_d The distribution factor accounts for the fact that the coils of one phase are distributed across several slots within a pole-pair rather than being concentrated in a single slot. The voltages induced in these spatially separated coils are not in phase, leading to a vector sum that is less than their arithmetic sum.

The factor is calculated using the slots per pole per phase, q, and the electrical angle between adjacent slots, α :

$$k_d = \frac{\sin\left(\frac{q\alpha}{2}\right)}{q\sin\left(\frac{\alpha}{2}\right)} \tag{2.5}$$

The electrical slot angle α is defined as the angular displacement between two adjacent slots, expressed in electrical degrees. For a three-phase motor ($m = 3$), it is calculated from the total slots per pole :

$$\alpha = \frac{180°}{\text{slots per pole}} = \frac{180°}{q \times m} = \frac{180°}{3q} \tag{2.6}$$

Pitch Factor, k_p The pitch factor accounts for the effect of *coil pitching* or *chording*, where the coil span (the distance between the two sides of a coil) is made shorter than the pole pitch (180 electrical degrees). While this slightly reduces the fundamental component, it significantly reduces specific space harmonics in the MMF waveform, leading to a smoother torque production and lower losses.

The factor is calculated from the coil span in electrical degrees, β :

$$k_p = \sin\left(\frac{\beta}{2}\right) \tag{2.7}$$

The coil span β is derived from the user-specified coil pitch percentage :

$$\beta = \frac{\text{Coil Pitch (\%)}}{100} \times 180^\circ \tag{2.8}$$

For example, a `5/6` pitch (83.33%) corresponds to a coil span of $\beta = 150^\circ$ and $k_p = \sin(75^\circ) \approx 0.966$.

Calculation Workflow in the Application

The calculation of K_w follows a deterministic sequence based on user inputs and derived parameters from other stages of the motor design process.

1. **Input Parameters :**
   — The number of poles ($2p$) and power rating are used with an internal recommendation table to determine the slots per pole per phase q.
   — The total number of stator slots Q is then derived : $Q = 2p \times m \times q$.
   — The user selects the **Winding Type** and **Coil Pitch** from drop-down menus (e.g., `Distributed`, `5/6 Pitch`).
2. **Calculate Slot Angle (α) :** Using Equation 3.8, the electrical angle between slots is computed. For a 4-pole motor with $q = 3$ and $Q = 36$ slots, $\alpha = 180^\circ/(3 \times 3) = 20^\circ$.
3. **Calculate Distribution Factor (k_d) :** Using the obtained q and α in Equation 2.5. For the example above : $k_d = \sin(3 \times 20^\circ/2)/(3 \times \sin(20^\circ/2)) \approx 0.960$.
4. **Calculate Pitch Factor (k_p) :** The coil pitch percentage is converted to an electrical span β using Equation 2.8. The pitch factor is then computed via Equation 2.7. For a `5/6` pitch : $k_p = \sin(150^\circ/2) \approx 0.966$.
5. **Compute Total Winding Factor (K_w) :** Finally, K_w is found by multiplying the two factors as per Equation 2.4 : $K_w = 0.960 \times 0.966 \approx 0.927$.

2.3 Module 2 : Electrical Sizing (Voltage, Current, Power)

The algorithm converts the mechanical output requirements specified by the user (Rated Power, Speed) into the electrical input parameters required to size the stator winding.

2.3.1 Algorithm Overview (Program Steps)

1. Determine the phase voltage E_{ph} from connection type (Y or Δ).
2. Estimate the electrical input power using $P_{elec} = \frac{P_{rated}}{\eta}$.
3. Compute the stator phase current I_s using the three-phase power relation with $\cos\varphi$.

2.3.2 Symbols and Parameters

— P_{rated} : rated mechanical output power [kW]
— P_{elec} : estimated electrical input power [kW]
— V_{line} : line-to-line supply voltage [V]
— E_{ph} : phase voltage [V]
— I_s (or I_{ph}) : stator phase current [A]
— η : efficiency (from Module 1) [-]
— $\cos\varphi$: power factor (from Module 1) [-]

2.3.3 1. Synchronous Speed (n_s)

The speed of the rotating magnetic field depends on the grid frequency (f) and the number of magnetic poles ($2p$).

$$p = \frac{\text{Poles}}{2}, \quad n_s = \frac{60f}{p} \quad [\text{RPM}] \tag{2.9}$$

The speed in revolutions per second (rps), often used in sizing equations, is simply $n_{s(rps)} = n_s/60$.

2.3.4 2. Electrical Input Power (P_{elec})

The motor must consume more electrical power than it produces in mechanical work due to internal losses (friction, windage, copper, and iron losses). The required electrical input power is calculated using the efficiency (η) estimated in Module 1 :

$$P_{elec} = \frac{P_{rated}}{\eta} \quad [kW] \tag{2.10}$$

Physical Significance : P_{elec} represents the total active power drawn from the supply. This value is higher than the nameplate power (P_{rated}) and determines the thermal load on the stator winding.

2.3.5 3. Phase Voltage and Current

To design the winding, the algorithm must determine the current flowing through a single phase conductor.

Phase Voltage (E_{ph})

The voltage across a phase winding depends on the stator connection type :
— **Star (Y) Connection :** $E_{ph} = V_{line}/\sqrt{3}$
— **Delta (Δ) Connection :** $E_{ph} = V_{line}$

Phase Current (I_s)

The phase current is derived from the electrical power equation for a three-phase system.

$$I_s = \frac{P_{elec} \times 1000}{3 \cdot E_{ph} \cdot \cos\phi} \quad [A] \tag{2.11}$$

Where :
— $\cos\phi$: The Power Factor estimated in Module 1.

— The factor 1000 converts P_{elec} from kW to Watts.

This current I_s is the critical parameter that dictates the required cross-sectional area of the copper conductors in Module 5.

2.4 Module 3 : Main Dimensions Synthesis (D and L)

This phase is the core of the geometric synthesis. It transitions from the abstract electrical requirements (Power, Speed) to the physical envelope of the machine. The process determines the Stator Bore Diameter (D) and the machine Length (L).

2.4.1 Algorithm Overview (Program Steps)

1. Compute the output coefficient C_o from B_{av}, q, K_w, η, and $\cos\varphi$.
2. Compute the required volume product D^2L from rated power and synchronous speed.
3. Select an aspect ratio λ (heuristic or preference-based) and solve for D and L.
4. Enforce the peripheral-speed constraint ; if violated, clamp D and recompute L.
5. Compute cooling ducts, stacking factor k_i, and the net iron length L_i.

2.4.2 Symbols and Parameters

— D : stator bore diameter (inner diameter) [m]
— L : gross axial length [m]
— D^2L : volume product of the active part [m^3]
— C_o : output coefficient [kW/(m^3 rps)]
— λ : aspect ratio $\lambda = \frac{L}{\tau_p}$ [-]
— τ_p : pole pitch $\tau_p = \frac{\pi D}{P}$ [m]
— v : peripheral speed [m/s]
— L_i : effective iron length (net stack length) [m]
— k_i : stacking factor [-]

The design logic relies on the **Output Equation**, which fundamentally relates the volume of the machine's active part (D^2L) to its power capability and speed.

2.4.3 The Output Coefficient (C_o)

The Output Coefficient is a composite metric representing the specific energy density of the machine. It combines magnetic loading, electric loading, and efficiency factors. It is calculated as :

$$C_o = 11 \cdot B_{av} \cdot q \cdot K_w \cdot \eta \cdot \cos\phi \cdot 10^{-3} \tag{2.12}$$

Where :
— B_{av} : Specific Magnetic Loading (Flux density in the air gap) [T].
— q : Specific Electric Loading (Ampere-conductors per meter) [A/m].
— K_w : Winding Factor (typically 0.955 for standard distribution).
— η : Target Efficiency.
— $\cos\phi$: Target Power Factor.

2.4.4 The Volume Product (D^2L)

With the specific loading (C_o) determined, the required volume of the stator bore is derived from the rated power (P) and synchronous speed (n_s in rps) :

$$D^2L = \frac{P_{kW}}{C_o \cdot n_{s(rps)}} \quad [m^3] \tag{2.13}$$

This equation provides the product of the dimensions but not the dimensions themselves. Infinite combinations of D and L can satisfy this volume. To separate them, the algorithm employs a shape factor known as the **Aspect Ratio** (λ).

2.4.5 Separation of Dimensions using Aspect Ratio (λ)

The Aspect Ratio λ is defined as the ratio of the stack length to the pole pitch (τ_p) :

$$\lambda = \frac{L}{\tau_p} \quad \text{where} \quad \tau_p = \frac{\pi D}{P_{poles}} \tag{2.14}$$

Substituting $L = \lambda \cdot \tau_p$ into the D^2L equation allows the software to solve for the diameter D explicitly :

$$D = \sqrt[3]{\frac{D^2L \cdot P_{poles}}{\pi \cdot \lambda}} \tag{2.15}$$

Once D is found, L is resolved simply as $L = D^2L/D^2$.

Strategies for Selecting λ

The software implements two distinct strategies to determine the optimal λ. The user's configuration determines which path is executed.

Method A : Heuristic for Best Power Factor (Fallback) If no specific constraints are applied, the software uses an empirical relationship designed to maximize the power factor. This method does not define λ directly but solves for L and D using fixed coefficients derived from historical best practices :

$$L = \sqrt{\frac{D^2L}{0.018225 \cdot P_{poles}^2}} \tag{2.16}$$

$$D = 0.135 \cdot P_{poles} \cdot \sqrt{L} \tag{2.17}$$

Method B : Preference-Based Optimization This is the primary method used for custom designs. The user selects a design goal (Cost, Efficiency, or Balance), and the software selects λ from specific ranges.

- **Minimum Cost ($\lambda \in [1.5 - 2.0]$) :** A higher λ yields a long, slender motor. This reduces the diameter, which minimizes the volume of the inactive "end-turns" of copper, significantly reducing material costs.
- **Good Efficiency ($\lambda \in [1.4 - 1.6]$) :** A moderate range that balances the resistive losses in the copper (which favor high λ) with the core losses in the iron (which favor low λ).
- **Balanced Design ($\lambda \in [1.0 - 1.1]$) :** A nearly square pole face, often used for general-purpose motors where neither cost nor efficiency is the sole driver.

2.4.6 Peripheral Speed Constraint Check

After calculating the initial Diameter (D_{calc}), the algorithm verifies mechanical safety by calculating the peripheral speed of the rotor surface (v) :

$$v = \pi \cdot D_{calc} \cdot n_{s(rps)} \tag{2.18}$$

Constraint : The standard limit is set to **30 m/s**.

— **Condition :** If $v \leq 30$, the calculated D_{calc} and L_{calc} are accepted.

— **Violation :** If $v > 30$, the diameter is clamped to the maximum safe limit :

$$D_{final} = \frac{30}{\pi \cdot n_{s(rps)}} \tag{2.19}$$

The length is then recalculated to preserve the required torque volume :

$$L_{final} = \frac{D^2 L}{D_{final}^2} \tag{2.20}$$

2.4.7 Effective Iron Length Calculation

The physical machine length (L) calculated above includes air and insulation. To find the magnetic active length (L_i), corrections are applied.

Radial Cooling Ducts

The number of radial cooling ducts (n_{ducts}) is specified as a user input. The duct width is fixed at 10 mm.

— **Duct Width (b_{duct}) :** Fixed at 10 mm.

— **Effective stack length (L_{s}) :** The total length of magnetic material is :

$$L_{\text{s}} = L - (n_{\text{ducts}} \times 0.010) \quad [\text{m}] \tag{2.21}$$

where L is the total stack length including ducts.

Iron Stacking Factor (k_i)

Laminations are insulated from each other, meaning the stack is not 100% steel. The stacking factor (k_i) accounts for this. Table 2.2 lists the lamination thickness associated with each steel grade used by the software.

TABLE 2.2 – Electrical steel grades and lamination thickness (used for stacking factor selection)

Grade	Thickness (mm)
M800-65A	0.65
M600-50A	0.50
M400-50A	0.50
M270-35A	**0.35**
M235-35A	0.35

— For standard laminations (0.50 mm thickness), $k_i \approx 0.95$.

— For thinner laminations (0.35 mm thickness), $k_i \approx 0.93$.

The final **Net Iron Length (L_i)**, used for all magnetic flux calculations, is :

$$L_i = L_s \cdot k_i \tag{2.22}$$

2.5 Module 4 : Winding Synthesis and Integer Constraints

With the physical envelope (D and L) established, the design process moves to the electromagnetic coils. The objective of this phase is to determine the precise number of wire turns required to induce the rated voltage (E_{ph}) without exceeding the magnetic limits of the steel core.

2.5.1 Algorithm Overview (Program Steps)

1. Use the EMF equation to compute a real-valued $T_{ph(calc)}$ from E_{ph}, f, Φ, and K_w.
2. Apply the integer constraint : set $T_{ph} = \lceil T_{ph(calc)} \rceil$.
3. Select the slots-per-pole-per-phase q from empirical recommendations and compute S_s.
4. Compute conductors per slot $N_{c(calc)}$ and apply integer rounding to obtain N_c.
5. Recompute manufacturing-ready winding totals (Z_s and $T_{ph(final)}$).
6. Back-calculate the flux required by the rounded winding and rescale length L ; recompute L_i.

2.5.2 Symbols and Parameters

— E_{ph} : phase voltage [V]
— f : supply frequency [Hz]
— Φ : flux per pole [Wb]
— T_{ph} : turns per phase [turns]
— K_w : winding factor [-]
— q : slots per pole per phase [-]
— S_s : number of stator slots [-]
— N_c : conductors per slot [-]
— Z_s : total conductors per phase [-]
— L_i : effective iron length (passed to slot sizing) [m]

2.5.3 Fundamental Concepts

— **Flux per Pole (Φ) :** The total amount of magnetic field generated by one magnetic pole. It is a function of the specific magnetic loading (B_{av}) and the pole area ($\tau_p \cdot L$).
— **Turns per Phase (T_{ph}) :** The total number of wire loops connected in series for one electrical phase.
— **Winding Factor (K_w) :** A coefficient (< 1) that accounts for the fact that coils are distributed across slots rather than concentrated in a single point, slightly reducing the effective voltage.

2.5.4 Calculating Turns per Phase (T_{ph})

The software utilizes the fundamental Electro-Motive Force (EMF) equation, which relates the changing magnetic field to the induced voltage (Faraday's Law) :

$$E_{ph} = 4.44 \cdot f \cdot \Phi \cdot T_{ph} \cdot K_w \tag{2.23}$$

Rearranging this to solve for the required turns :

$$T_{ph(calc)} = \frac{E_{ph}}{4.44 \cdot f \cdot \Phi_{initial} \cdot K_w} \tag{2.24}$$

The Integer Constraint : The result of this calculation is a real number (e.g., 82.34 turns). However, it is physically impossible to manufacture a fraction of a turn. The software applies a strict rounding rule :

$$T_{ph} = \lceil T_{ph(calc)} \rceil \quad \text{(Rounded Up)} \tag{2.25}$$

The code forces the value to the next highest integer. This seemingly minor adjustment is critical because increasing the turns (T_{ph}) changes the magnetic balance of the machine.

2.5.5 Stator Slot Distribution

Once the total turns are known, they must be distributed into slots around the stator.

Slots per Pole per Phase (q)

This parameter determines how "distributed" the winding is. A higher q produces a smoother sinusoidal wave but increases manufacturing complexity. The software selects q from empirical lookup tables (Table 7.5 in the codebase) based on the machine's power and pole count.

— Typical values : $q = 2, 3, 4$, or 5.

TABLE 2.3 – Recommended Slots per Pole per Phase (q) (Table 7.5)

Power Range	**Recommended q by Pole Count ($2p$)**			
(kW)	**2 Poles**	**4 Poles**	**6 Poles**	**8 Poles**
$0.5 - 5$	$2 - 3$	$2 - 3$	2	$1 - 2$
$5 - 50$	$3 - 4$	3	$2 - 3$	2
$50 - 200$	$4 - 5$	$3 - 4$	3	$2 - 3$
$200 - 1000$	$5 - 6$	$4 - 5$	$3 - 4$	$3 - 4$

Total Slots (S_s)

The total number of slots stamped into the stator lamination is :

$$S_s = 2p \cdot m \cdot q \tag{2.26}$$

Where $2p$ is the number of poles and $m = 3$ is the number of phases.

Conductors per Slot (N_c)

The software calculates how many electrical conductors must fit inside a single slot :

$$N_{c(calc)} = \frac{T_{ph} \cdot 2 \cdot m}{S_s} \tag{2.27}$$

Note : The factor of 2 accounts for the two "sides" of a coil loop.

The Second Integer Constraint : Like turns, N_c must be an integer. The software rounds $N_{c(calc)}$ up to the nearest integer. This rounding may slightly increase the total

number of turns again. The final, manufacturing-ready winding parameters are recalculated as :

$$Z_s = N_c \cdot \frac{S_s}{3} \quad \text{(Total Conductors per Phase)} \tag{2.28}$$

$$T_{ph(final)} = \frac{Z_s}{2} \quad \text{(Final Turns per Phase)} \tag{2.29}$$

2.5.6 Algorithmic Feedback Loop : The Geometry Adjustment

Because T_{ph} was rounded up, the machine now has more turns than the initial EMF calculation required. If the physical dimensions (D and L) remained unchanged, the machine would operate at a lower magnetic flux density than intended, effectively "under-utilizing" the steel core.

To optimize the design, the software performs a **Back-Calculation** to adjust the physical machine length (L).

Step 1 : Recalculate Flux

Using the final integer T_{ph}, the software solves the EMF equation in reverse to find the actual Flux per Pole required to meet the voltage rating :

$$\Phi_{new} = \frac{E_{ph}}{4.44 \cdot f \cdot K_w \cdot T_{ph(final)}} \tag{2.30}$$

Step 2 : Scaling the machine Length (L_{new})

Since Flux is proportional to area ($\Phi = B \cdot \text{Area} \propto L$), the machine length is scaled by the ratio of the new flux to the old flux :

$$\text{Ratio} = \frac{\Phi_{new}}{\Phi_{initial}} \tag{2.31}$$

$$L_{new} = L_{initial} \times \text{Ratio} \tag{2.32}$$

Physical Significance :

- If rounding T_{ph} up resulted in a significantly higher number of turns, Φ_{new} will decrease.
- Consequently, L_{new} will decrease.
- **Result :** The algorithm automatically shortens the motor to save material (steel) while maintaining the exact required electromagnetic performance.

2.5.7 Finalizing the Iron Length

With the new physical length L_{new} established, the cooling duct calculation (Module 3) is re-run to determine the final Net Iron Length (L_i) :

$$L_s = L_{new} - (n_{ducts} \cdot b_{duct}) \tag{2.33}$$

$$L_i = L_s \cdot k_i \tag{2.34}$$

This L_i is the value passed forward to the Slot Design phase for calculating tooth flux densities.

2.6 Slot Design Philosophy and Industry Alignment

2.6.1 Slot Shape Impact on Machine Performance

The shape of stator slots significantly impacts both the manufacturing process and electromagnetic performance of induction machines. Figure 2.1 shows three prevalent slot configurations :

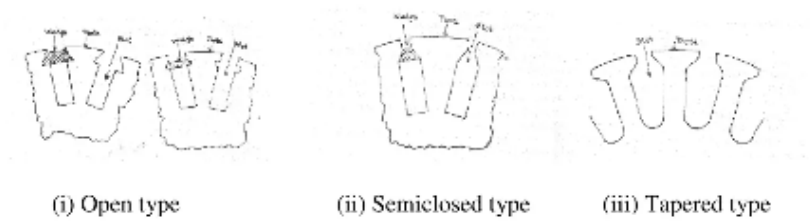

FIGURE 2.1 – Typical stator slot shapes : (i) Open type, (ii) Semiclosed type, and (iii) Tapered type.

— **Open slots** facilitate easy insertion of pre-formed coils but may increase air-gap flux harmonics and cogging torque (mainly used for high voltage/high power machines).
— **Semiclosed slots** reduce flux leakage and are preferred for distributed windings in small to medium motors.
— **Tapered slots** provide better utilization of the tooth cross-section for magnetic flux while accommodating winding space requirements.

2.6.2 Pragmatic Approach : Simplified Slot-Shape Assignment

Given these options, the application implements a **pragmatic, industry-aligned approach** to slot shape selection based on voltage class and manufacturing constraints, rather than offering all theoretical possibilities. This simplification reflects real-world induction motor manufacturing practices where slot geometry is primarily dictated by winding technology and insulation requirements.

2.6.3 Two-Tier Design Strategy

TABLE 2.4 – Slot Shape Assignment Strategy

Low Voltage (LV) Machines **(V 600V, Random-Wound)**	**High Voltage/High Power (HV/HP) Machines** **(V > 600V, Form-Wound)**
Slot Shape : Trapezoidal (Semi-Closed)	**Slot Shape : Rectangular (Open Slot)**
• Uses **round enameled wire** inserted conductor-by-conductor • Narrow slot opening prevents coil movement • Better magnetic performance (reduced slot harmonics) • Standard for IEC frame motors up to 375 kW (500 Hp) • Maximizes copper fill factor within slot	• Uses **pre-formed rectangular conductors** with insulation • Wide opening allows insertion of rigid coils • Manufacturing necessity for form-wound windings • Standard for motors above 375 kW (500 Hp) / 600 V • Accommodates thick ground wall insulation

2.6.4 Why Only One Shape Per Category ?

For LV Machines : Why Trapezoidal Prevails

The trapezoidal slot with parallel-sided teeth represents the **electromagnetic optimum** for efficiency-focused designs. As established by Pyrhönen et al. in *Design of Rotating Electrical Machines*, this shape :

- Maximizes slot area for a given tooth width at the air-gap
- Achieves the highest copper fill factor (reducing I^2R losses)
- Provides adequate mechanical support for random-wound conductors
- Offers the best compromise between copper loss (dominant in LV motors) and iron loss

While rectangular slots are possible for LV machines, they are less common in performance-oriented designs due to increased slot harmonics and slightly lower fill factors.

For HV/HP Machines : Why Rectangular is Mandatory

High-voltage machines (typically >600 V) require form-wound coils with multi-layer insulation systems. The **open rectangular slot** becomes a manufacturing imperative because :

- Pre-insulated rigid coils cannot be inserted through narrow openings
- Slot liners and wedges require straight-sided geometry
- The electromagnetic disadvantages (increased Carter factor, stray losses) are mitigated through complementary design features :
  - Magnetic slot wedges to smooth air-gap flux
  - Rotor slot skewing to reduce harmonics
  - Optimized slot/pole combinations

2.6.5 Design Focus and Validation

By limiting to one shape per category, the application :

1. Focuses computational resources on optimizing the **most relevant parameters** (dimensions, winding, materials) rather than exploring impractical geometries
2. Ensures designs are **manufacturable** from the outset by respecting industry-standard winding techniques
3. Aligns with **industry standards** (IEC, NEMA) where slot shapes are typically predetermined by voltage class
4. Reduces user confusion by eliminating impractical combinations (e.g., form-wound coils in tapered slots)

2.7 Module 5 : Conductor and Slot Sizing

This phase transforms the electrical requirements obtained previously into practical conductor choices and slot geometries. As established in Section [ref to new section], the application employs a **two-tier strategy** based on voltage class :

— **Low Voltage (LV) Machines :** Trapezoidal slot with round conductors (random-wound)

— **High Voltage/High Power (HV/HP) Machines :** Rectangular open slot with rectangular strips (form-wound)

The following sections detail the specific algorithms for each category, reflecting industry-standard design practices.

2.7.1 Algorithm Overview (Program Steps)

1. Determine the slot/conductor method from user selection and from LV/HV-HP criteria (wire vs strip).
2. Compute the required copper area from $a_c = \frac{I_{ph}}{J}$, where J is obtained from an abaque.
3. Standardize the conductor : (LV) select number of strands and SWG so that $d \leq 1.7$ mm ; (HV/HP) select strip dimensions from the open-slot abaque.
4. Compute required slot area using $A_{slot,req} = \frac{T_{ac}}{k_{fill}}$ with the method-specific fill factor.
5. Apply tooth saturation sizing (compute ϕ_{max} and a minimum tooth width) and solve slot dimensions with the chosen geometric model.

2.7.2 Symbols and Parameters

— D : stator bore diameter (inner diameter)

— S_s : number of stator slots

— P : number of poles

— L_i : effective iron length (net stack length)

— I_{ph} (or I_s) : stator phase current

— J : admissible/recommended current density (from an abaque)

— a_c : required copper area for one conductor, $a_c = I_{\text{ph}}/J$

— $a_{c,\text{std}}$: actual copper area after selecting a standard wire/strip

— N_c : conductors per slot

— T_{ac} : total copper area per slot, $T_{ac} = a_{c,\text{std}}\, N_c$

— k_{fill} : slot fill factor (typically 0.4 for LV random-wound ; 0.7 for HV/HP formed conductors)

— $A_{\text{slot,req}}$: required slot area, $A_{\text{slot,req}} = T_{ac}/k_{\text{fill}}$

— $\tau_s(D)$: slot pitch at diameter D, $\tau_s(D) = \pi D/S_s$
— Φ_{pole} : flux per pole
— $\phi_{\max}$: maximum tooth flux used for tooth-width sizing
— $B_{\text{tooth,max}}$: maximum allowable tooth flux density (steel/design limit)
— w_{st1} : tooth width at bore (minimum from saturation constraint)
— b_s, h_s : open-slot rectangular approximation (slot width and slot height)

2.7.3 Abaques (Reference Tables) Used in Module 5

These reference tables are shared by the different slot/conductor methods below. When the input lies between two tabulated points, the software uses linear interpolation.

Electrical Steel : Max Design Flux Density for Tooth Sizing

The tooth minimum width is constrained by a maximum admissible tooth flux density $B_{\text{tooth,max}}$, selected from the chosen steel grade.

Table 2.5 – Electrical steel grades and maximum design flux density (used in tooth-width sizing)

Grade	**Max Design B (T)**
M800-65A	1.55
M600-50A	1.60
M400-50A	1.65
M270-35A	**1.70**
M235-35A	1.75

Current Density Table $J(D)$

Table 2.6 – Recommended current density J vs bore diameter D

D (mm)	100	150	200	300	400	500	750	1000
J (A/mm^2)	4.0	3.8	3.6	3.5	3.5	3.5	3.5	3.5

Standard Wire Gauge (SWG) Excerpt (LV random-wound)

The LV random-wound methods select a standard wire diameter from an internal SWG table. Table 2.7 shows an excerpt.

Table 2.7 – Excerpt of SWG Wire Table Used

SWG	**Diameter (mm)**	**SWG**	**Diameter (mm)**
16	1.626	21	0.813
17	1.422	22	0.711
18	1.219	23	0.610
19	1.016	24	0.559
20	0.914	25	0.508

Open-Slot Strip Abaque (HV/HP formed conductors)

TABLE 2.8 – Open-slot abaque : recommended strip dimensions versus phase current

$I_{\min}$ (A)	$I_{\max}$ (A)	$t_{\min}$ (mm)	$t_{\max}$ (mm)	$w_{\min}$ (mm)	$w_{\max}$ (mm)
0	20	1.0	1.5	2.5	5.0
20	40	1.5	2.0	4.0	7.0
40	80	1.9	2.5	6.0	9.0
80	150	2.5	3.2	8.0	11.0
150	∞	3.2	4.0	10.0	14.0

TABLE 2.9 – Open-slot abaque : strip aspect ratio range (width/thickness)

Current row	$(w/t)_{\min}$	$(w/t)_{\max}$
0–20 A	2.5	3.5
20–40 A	2.5	3.5
40–80 A	2.8	3.5
80–150 A	3.0	3.5
150 A–∞	3.0	3.5

TABLE 2.10 – Standard strip thicknesses

Thickness values (mm)											
1.00	1.25	1.50	1.60	1.80	1.90	2.00	2.24	2.50	2.80	3.15	3.55
4.00											

2.7.4 How to determine the maximum flux the tooth should carry

This is a detailed step-by-step of derivation of the maximum magnetic flux in a stator tooth (the flux value that theoretically the tooth should carry) of a rotating electrical machine (induction motors for our case). This calculation is essential for proper magnetic design, ensuring that the tooth cross-sectional area is sufficient to prevent magnetic saturation. The derivation follows the classical approach : assuming a sinusoidal air-gap flux density distribution, integrating over the tooth's catchment area, and relating the result to the machine's overall magnetic loading.

Physical Setup and Notation

Consider a **stator** with the following parameters :
— N_s : Number of stator slots
— P : Number of poles
— D : Stator bore diameter
— L : Axial length of the machine (stack length)
— B_{av} : Average flux density over a pole pitch
— $B_{\max}$: Peak value of the sinusoidal air-gap flux density
— Φ_{pole} : Magnetic flux per pole

Sinusoidal Air-Gap Flux Density Distribution

Let θ be the **mechanical angle** measured along the stator bore. The radial flux density in the air gap is modeled as :

$$B_r(\theta) = B_{\max} \cos\left(\frac{P}{2}\theta\right) \tag{2.35}$$

The argument $\frac{P}{2}\theta$ converts mechanical angle to **electrical radians**. The cosine function is chosen here so that its maximum occurs at $\theta = 0$. We will position our coordinate system such that the center of the tooth under investigation is at $\theta = 0$, meaning the peak of the flux density wave is aligned with that tooth center to obtain the *maximum possible flux* for that tooth.

Why a sine wave ? In an ideal machine with sinusoidally distributed windings and a sinusoidal current excitation, the fundamental space harmonic of the air-gap field is sinusoidal. Higher space harmonics are neglected in this basic sizing calculation.

Slot Pitch and Tooth Catchment Area

Slot Pitch The **slot pitch** τ_s is the angular distance from the center of one slot to the center of the next slot :

$$\tau_s = \frac{2\pi}{N_s} \quad \text{(in mechanical radians)} \tag{2.36}$$

Concept of Catchment Area Magnetic flux enters the stator from the air gap. Each tooth is responsible for collecting the flux from a certain portion of the stator's inner surface. A standard design assumption is that the flux entering over an angular span of **one slot pitch** is collected by a single tooth. This means the tooth gathers flux from half the slot on its left and half the slot on its right.

Therefore, if we place the center of the tooth at $\theta = 0$, the angular boundaries of its catchment area are :

$$\theta = -\frac{\tau_s}{2} \quad \text{to} \quad \theta = +\frac{\tau_s}{2}$$

where $\tau_s/2$ is the **half slot pitch**.

Flux Calculation via Integration

Flux through an Infinitesimal Arc Consider a small segment of the stator bore surface at angle θ, subtending an angle $d\theta$. Its area is :

$$dA = (\text{arc length}) \times (\text{stack length}) = \left(\frac{D}{2}d\theta\right) \times L = \frac{DL}{2}\,d\theta$$

The infinitesimal magnetic flux through this area is :

$$d\Phi = B_r(\theta)\,dA = B_r(\theta)\,\frac{DL}{2}\,d\theta$$

Integration over the Tooth Catchment Area The total flux Φ_{tooth} entering the tooth is the integral of $d\Phi$ over its catchment area :

$$\Phi_{\text{tooth}} = \int_{-\tau_s/2}^{\tau_s/2} d\Phi = \frac{DL}{2}\int_{-\tau_s/2}^{\tau_s/2} B_r(\theta)\,d\theta \tag{2.37}$$

Substituting Eq. (2.35) and $\tau_s = 2\pi/N_s$:

$$\Phi_{\text{tooth}} = \frac{DL}{2} B_{\max} \int_{-\pi/N_s}^{\pi/N_s} \cos\left(\frac{P}{2}\theta\right) d\theta$$

Performing the Integration The integral of $\cos(k\theta)$ is $\frac{1}{k}\sin(k\theta)$, with $k = P/2$. Thus :

$$\begin{aligned}
\int \cos\left(\frac{P}{2}\theta\right) d\theta &= \frac{2}{P}\sin\left(\frac{P}{2}\theta\right) \\
\int_{-\pi/N_s}^{\pi/N_s} \cos\left(\frac{P}{2}\theta\right) d\theta &= \frac{2}{P}\left[\sin\left(\frac{P}{2}\cdot\frac{\pi}{N_s}\right) - \sin\left(\frac{P}{2}\cdot\left(-\frac{\pi}{N_s}\right)\right)\right] \\
&= \frac{2}{P}\left[\sin\left(\frac{P\pi}{2N_s}\right) - \sin\left(-\frac{P\pi}{2N_s}\right)\right]
\end{aligned}$$

Since $\sin(-x) = -\sin(x)$, the expression simplifies to :

$$\int_{-\pi/N_s}^{\pi/N_s} \cos\left(\frac{P}{2}\theta\right) d\theta = \frac{2}{P} \times 2\sin\left(\frac{P\pi}{2N_s}\right) = \frac{4}{P}\sin\left(\frac{P\pi}{2N_s}\right)$$

Substituting Back Insert this result into Eq. (2.37) :

$$\begin{aligned}
\Phi_{\text{tooth}} &= \frac{DL}{2} B_{\max} \cdot \frac{4}{P}\sin\left(\frac{P\pi}{2N_s}\right) \\
&= \frac{2DLB_{\max}}{P}\sin\left(\frac{P\pi}{2N_s}\right) \qquad (2.38)
\end{aligned}$$

Relation to Flux per Pole and Average Flux Density

Flux per Pole Φ_{pole} The **flux per pole** is the total flux emanating from one magnetic North pole. It is obtained by integrating $B_r(\theta)$ over one **pole pitch** $\tau_p = 2\pi/P$. For the sinusoidal distribution, integrate from $-\tau_p/2$ to $+\tau_p/2$:

$$\begin{aligned}
\Phi_{\text{pole}} &= \int_{-\tau_p/2}^{\tau_p/2} B_{\max}\cos\left(\frac{P}{2}\theta\right)\frac{DL}{2} d\theta \\
&= \frac{DL}{2} B_{\max} \cdot \frac{4}{P}\sin\left(\frac{P}{2}\cdot\frac{\tau_p}{2}\right)
\end{aligned}$$

Since $\tau_p = 2\pi/P$, we have $\frac{P}{2}\cdot\frac{\tau_p}{2} = \frac{P}{2}\cdot\frac{\pi}{P} = \frac{\pi}{2}$, and $\sin(\pi/2) = 1$. Therefore :

$$\Phi_{\text{pole}} = \frac{2DLB_{\max}}{P} \qquad (2.39)$$

Notice that the factor $\frac{2DLB_{\max}}{P}$ appears in our interim result for Φ_{tooth} (Eq. (2.38)).

Average Flux Density B_{av} The **average flux density** over a pole pitch is defined as the flux per pole divided by the area of the pole face :

$$\text{Pole face area} = (\text{pole arc length}) \times L = \left(\frac{\pi D}{P}\right) L = \frac{\pi DL}{P}$$

Thus :

$$B_{\text{av}} = \frac{\Phi_{\text{pole}}}{\pi DL/P} = \frac{\frac{2DLB_{\max}}{P}}{\frac{\pi DL}{P}} = \frac{2}{\pi} B_{\max} \qquad (2.40)$$

Or equivalently :

$$B_{\max} = \frac{\pi}{2} B_{\text{av}}$$

Physical meaning of B_{av} : $B_{\max}$ is the peak of the sinusoidal spatial wave. B_{av} is the equivalent uniform flux density that would produce the same total flux per pole. It is a practical design parameter, often chosen based on experience and material properties.

We can also express Φ_{pole} in terms of B_{av} :

$$\Phi_{\text{pole}} = B_{\text{av}} \times (\text{pole face area}) = B_{\text{av}} \cdot \frac{\pi D L}{P} \tag{2.41}$$

Final Expression for Maximum Tooth Flux

Comparing Eq. (2.38) and Eq. (2.39), we see that :

$$\boxed{\Phi_{\text{tooth,max}} = \Phi_{\text{pole}} \cdot \sin\left(\frac{P\pi}{2N_s}\right)} \tag{2.42}$$

This is the key result. Since we aligned the peak of the flux density wave with the tooth center, this gives the *maximum possible flux* for any tooth. For teeth not aligned with the peak, the flux will be smaller.

Interpretation of the Factor $\sin\left(\frac{P\pi}{2N_s}\right)$

- The argument $\frac{P\pi}{2N_s} = \frac{P}{2} \cdot \frac{\pi}{N_s} = \frac{P}{2} \cdot \frac{\tau_s}{2}$. This is the **electrical angle** corresponding to **half a slot pitch**.
- This factor, always less than 1, represents the fraction of the pole flux that passes through a single tooth when the tooth is at the most favorable position.
- If the number of slots per pole N_s/P is large, this angle is small, $\sin(x) \approx x$, and the tooth flux is only a small fraction of the pole flux.
- The factor depends on the machine's **slots per pole per phase** geometry and influences the tooth width design.

Application in Machine Design

The maximum tooth flux $\Phi_{\text{tooth,max}}$ is used to determine the required **tooth cross-sectional area** A_{tooth} :

$$A_{\text{tooth}} \approx w_{\text{tooth}} \times L \times k_{\text{Fe}}$$

where w_{tooth} is the tooth width (circumferential), and $k_{\text{Fe}} < 1$ is the stacking (lamination) factor.

To avoid magnetic saturation, the flux density in the tooth must remain below the saturation limit of the core material (typically 1.6–1.8 T for silicon steel) :

$$B_{\text{tooth}} = \frac{\Phi_{\text{tooth,max}}}{A_{\text{tooth}}} < B_{\text{sat}}$$

This inequality sets a lower bound for the tooth width :

$$w_{\text{tooth}} \geq \frac{\Phi_{\text{tooth,max}}}{L \cdot k_{\text{Fe}} \cdot B_{\text{sat}}}$$

Using Eq. (2.42) and Eq. (2.41), the designer can compute $\Phi_{\text{tooth,max}}$ from the chosen overall magnetic loading B_{av} and machine dimensions D, L, P, N_s.

2.7.5 Low Voltage Implementation : Trapezoidal Slot Design

This method is the primary design choice for standard low-voltage induction machines ($V \leq 600$V). It utilizes **Random Wound Coils** (typically round enameled wire) inserted through a narrow slot opening.

Design Philosophy : Constant Tooth Width

The geometric priority of this method is to maintain a **constant magnetic tooth width** (w_{st}) along the entire radial depth of the slot.

- **Advantage :** Ensures uniform magnetic flux density in the teeth, preventing localized saturation.
- **Consequence :** Because the stator radius increases as we move outward from the bore, maintaining a constant tooth width requires the slot to widen (taper) towards the yoke.

The detailed algorithmic implementation for the trapezoidal slot design covers conductor selection, geometric definitions, and area matching.

1. Conductor Selection (Random Wound)

For low-voltage applications, the software selects standard round wire rather than rectangular strips.

Current Density and Area The target copper area is determined by the phase current (I_s) and the allowable current density (J), which is interpolated based on the machine size.

$$a_{req} = \frac{I_s}{J} \quad [mm^2] \tag{2.43}$$

Standard Wire Gauge (SWG) Selection The software iterates through the SWG table to find a standard wire diameter (d_{std}) and number of parallel strands ($n_{strands}$) that satisfy the area requirement.

- **Constraint :** Max single wire diameter ≤ 1.7 mm (to ensure flexibility for random winding).
- **Selection Rule :** $n_{strands} \times \frac{\pi \cdot d_{std}^2}{4} \geq a_{req}$.

See Table 2.7 (Module 5 Abaques) for the SWG excerpt used.

2. Geometric Definitions

The slot shape is defined by fixed vertical regions and variable widths calculated to maintain the tooth profile.

- h_{s0} **(Opening Height) :** Fixed at **1.0 mm** (slot opening height).
- h_{s1} **(Wedge Height) :** Fixed at **3.0 mm** (tooth shoe height).
- b_{s0} **(Opening Width) :** $3 \times d_{std}$ (Sized for wire insertion).
- w_{st} **(Tooth Width) :** Calculated from magnetic flux limits : $w_{st} = \frac{\Phi_{max}}{B_{max} \cdot L_i}$.

3. Area Matching Algorithm

Since the slot is tapered, the active cross-sectional area (A_{slot}) is a function of the active depth (h_{s2}). The software employs a numerical solver to find the exact h_{s2} that accommodates the copper winding.

Step A : Calculate Top Width (b_{s1}) The width at the top of the active winding area (just below the wedge) is geometrically fixed by the bore diameter (D) and the required tooth width (w_{st}).

$$b_{s1} = 2\left[\tan\left(\frac{\pi}{S_s}\right)\left(\frac{D}{2} + h_{s0} + h_{s1}\right) - \frac{w_{st}/2}{\cos(\pi/S_s)}\right] \tag{2.44}$$

Step B : Iterative Depth Search The algorithm iterates to find the active depth (h_{s2}).

1. **Target Area :** $A_{target} = \frac{N_c \cdot a_{copper}}{k_{fill}}$ (where $k_{fill} \approx 0.4$).
2. **Initialize :** $h_{s2} = 0.1$ mm.
3. **Loop :**
   — Calculate Bottom Width (b_{s2}) at current depth :

$$b_{s2} = 2\left[\tan\left(\frac{\pi}{S_s}\right)\left(\frac{D}{2} + h_{s0} + h_{s1} + h_{s2}\right) - \frac{w_{st}/2}{\cos(\pi/S_s)}\right] \tag{2.45}$$

   — Calculate Area : $A_{calc} = \frac{(b_{s1}+b_{s2})\cdot h_{s2}}{2}$.
   — Check Error : If $\frac{|A_{calc}-A_{target}|}{A_{target}} > 2\%$, increase h_{s2}.

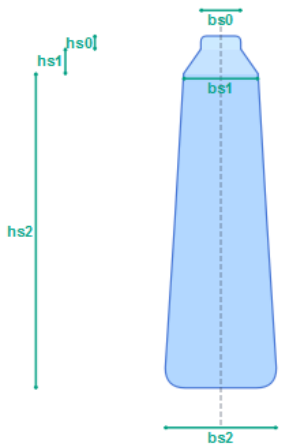

FIGURE 2.2 – Geometry details of tapered slot type (LV machines)

2.7.6 High Voltage Machine Slot Design : Open Slot

Scope and Typical Applications

This method is intended for **three-phase induction machines** in the **high-voltage (HV)** and/or **high-power (HP)** range, where the stator commonly uses **form-wound rectangular conductors** (stator bars / rectangular strips) and where an **open slot** is preferred for manufacturability (bar insertion, insulation build, wedge placement). The method couples :

— electromagnetic tooth saturation constraints (through a maximum allowable tooth flux density),

— geometric feasibility constraints (slot pitch and aspect ratio),
— copper accommodation constraints (slot area from conductor area and fill factor),
— practical conductor selection via empirical abaques (recommended strip dimensions vs phase current).

Abaques and Reference Tables Used

The HV/HP open-slot method uses the same shared Phase 5 reference tables : Table 2.6, Table 2.8, Table 2.9, and Table 2.10.

Methodology :

A. Electromagnetic tooth-flux constraint (defines w_{st1})

A1. Maximum tooth flux. Compute the maximum tooth flux using :

$$\phi_{\max} = \Phi_{\text{pole}} \sin\left(\frac{\pi P}{2S_s}\right). \tag{2.46}$$

A2. Minimum tooth width at bore. Select a maximum allowable tooth flux density $B_{\text{tooth,max}}$ (typically a material/design limit). Then :

$$w_{st1} = \frac{\phi_{\max}}{B_{\text{tooth,max}}\, L_i}. \tag{2.47}$$

This w_{st1} is the **minimum** tooth width at the bore required to keep tooth flux density at or below $B_{\text{tooth,max}}$.

B. Slot pitch constraint at the bore (defines a pitch-limited b_s) Slot pitch at the bore is :

$$\tau_s(D) = \frac{\pi D}{S_s}. \tag{2.48}$$

Assuming an open slot where the slot width is b_s at the bore, the pitch is split into tooth and slot :

$$\tau_s(D) = w_{st1} + b_s. \tag{2.49}$$

Hence the pitch-limited slot width is :

$$b_{s,\text{pitch}} = \tau_s(D) - w_{st1}. \tag{2.50}$$

C. Rectangular strip selection from abaques (defines A_{strip})

C1. Select current density J. Determine J from Table 2.6 by interpolation using the bore diameter D.

C2. Required conductor copper area. Compute the required copper cross-section for one conductor :

$$a_c = \frac{I_{\text{ph}}}{J}. \tag{2.51}$$

C3. Choose recommended ranges by current. Locate the abaque row in Table 2.8 such that :

$$I_{\min} \leq I_{\text{ph}} < I_{\max}.$$

This yields recommended ranges $t \in [t_{\min}, t_{\max}]$ and $w \in [w_{\min}, w_{\max}]$.

C4. Choose a standard thickness t_{std}. Pick t_{std} from Table 2.10 such that :

$$t_{\min} \leq t_{\text{std}} \leq t_{\max},$$

preferably the smallest available thickness in-range.

C5. Choose width w_{std} by stepping. Initialize w at the next 0.5 mm increment above $w_{\min}$, then increase by $\Delta w = 0.5$ mm until :

$$A_{\text{strip}} = t_{\text{std}}\, w_{\text{std}} \geq a_c, \tag{2.52}$$

or until w_{std} reaches $w_{\max}$ (in which case the closest feasible within-range choice is kept).

D. Slot area requirement from fill factor (defines $A_{\text{slot,req}}$) For HV/HP form-wound rectangular conductors, the method uses a higher fill factor :

$$k_{\text{fill}} = 0.7.$$

Compute the total copper to be placed in a slot :

$$T_{ac} = A_{\text{strip}}\, N_c. \tag{2.53}$$

Then the required slot area is :

$$A_{\text{slot,req}} = \frac{T_{ac}}{k_{\text{fill}}}. \tag{2.54}$$

E. Slot geometry solve (compute b_s and h_s) The geometric model is a simple open-slot rectangle where the **net slot area** is approximated by :

$$A_{\text{slot}} \approx b_s\, h_s. \tag{2.55}$$

A practical aspect ratio constraint is enforced :

$$3 \leq \frac{h_s}{b_s} \leq 5. \tag{2.56}$$

E1. Candidate A (pitch-fixed). Keep the pitch constraint strict by setting :

$$b_s = b_{s,\text{pitch}}.$$

Then choose the smallest h_s that satisfies both area and the lower aspect bound :

$$h_s = \max\left(\frac{A_{\text{slot,req}}}{b_s},\ 3b_s\right). \tag{2.57}$$

This candidate always satisfies :

$$A_{\text{slot}} \geq A_{\text{slot,req}}, \qquad \frac{h_s}{b_s} \geq 3,$$

but it may violate $\frac{h_s}{b_s} \leq 5$ if $A_{\text{slot,req}}$ is large or b_s is small.

E2. Candidate B (ratio-fixed at 5 with exact area). Force the extreme allowed aspect ratio and exact area :

$$\frac{h_s}{b_s} = 5, \qquad b_s h_s = A_{\text{slot,req}}. \tag{2.58}$$

Solving gives :

$$b_s = \sqrt{\frac{A_{\text{slot,req}}}{5}}, \qquad h_s = 5b_s. \tag{2.59}$$

Now check the bore pitch feasibility :

$$\Delta_{\text{pitch}} = (b_s + w_{st1}) - \tau_s(D). \tag{2.60}$$

If $\Delta_{\text{pitch}} \leq 0$, then $b_s + w_{st1} \leq \tau_s(D)$ and the pitch can be satisfied by **increasing** tooth width to :

$$w_{st1,\text{req}} = \tau_s(D) - b_s, \tag{2.61}$$

which **reduces** tooth flux density (safe). This is acceptable in HV/HP design because it keeps B_{tooth} below the limit.

If $\Delta_{\text{pitch}} > 0$, satisfying pitch would require **reducing** w_{st1}, which would increase tooth flux density and is not allowed ; in that case Candidate A is preferred even if the ratio exceeds 5.

F. Tooth width at slot top and flux-density check With the chosen slot height h_s and slot width b_s, compute pitch at the slot top (diameter $D + 2h_s$) :

$$\tau_s(D + 2h_s) = \frac{\pi(D + 2h_s)}{S_s}. \tag{2.62}$$

Then tooth width at the top of the slot is :

$$w_{st2} = \tau_s(D + 2h_s) - b_s. \tag{2.63}$$

Compute tooth flux density at bore and at slot top :

$$B_{\text{tooth,bore}} = \frac{\phi_{\max}}{L_i\, w_{st1}}, \qquad B_{\text{tooth,top}} = \frac{\phi_{\max}}{L_i\, w_{st2}}. \tag{2.64}$$

The bore value is the critical one by construction (it should be at or below $B_{\text{tooth,max}}$), while the top value is typically smaller because $w_{st2} > w_{st1}$ due to larger pitch at larger diameter.

Practical Notes

- The fill factor $k_{\text{fill}} = 0.7$ reflects HV/HP formed rectangular windings and is significantly higher than typical random-wound values.
- The constraint interval $3 \leq h_s/b_s \leq 5$ is a practical geometry rule to avoid excessively narrow/deep slots or overly wide/shallow slots.
- The abaque tables are empirical ; they should be validated against manufacturing constraints (minimum insulation thickness, wedge space, liner space, corner radii) if detailed mechanical design is required.

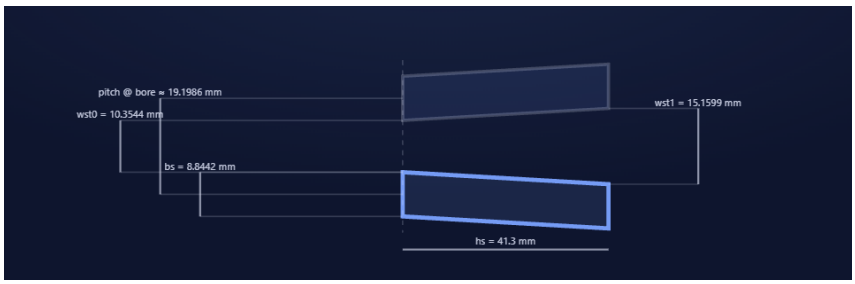

FIGURE 2.3 – Geometry details of open slot type (HV/HP machines)

2.8 Module 6 : Yoke Sizing and Outer Diameter Calculation

This phase sizes the **stator yoke (back iron)** so that it can carry the required flux without exceeding a chosen yoke flux-density level, then converts that yoke thickness into the **stator outer diameter**. The method is used as a late-stage geometric closure step after the main dimensions, flux, and slot height have been established.

2.8.1 Algorithm Overview (Program Steps)

1. Compute the yoke flux $\Phi_{yoke} = \frac{\Phi}{2}$ using the ring-yoke split assumption.
2. Select a yoke flux-density target B_{yoke} (kept below the tooth target).
3. Compute required yoke area $A_c = \frac{\Phi_{yoke}}{B_{yoke}}$ and thickness $d_{cs} = \frac{A_c}{L_i}$.
4. Compute $D_{ext,calc} = D + 2h_{slot} + 2d_{cs}$ and round to the nearest manufacturing increment.

2.8.2 Symbols and Parameters

— D : stator bore diameter (inner diameter)
— D_{ext} : stator outer diameter
— h_{slot} : total slot height (radial)
— d_{cs} : yoke (core/back-iron) radial thickness
— Φ : flux per pole
— Φ_{yoke} : flux circulating in the stator yoke
— B_{yoke} : yoke flux density used for sizing
— A_c : required yoke cross-sectional area
— L_i : effective iron length (net stack length)

2.8.3 Step 1 : Flux in the stator yoke

In a classical ring-yoke approximation, the pole flux splits into two parallel paths in the back iron. Therefore :

$$\Phi_{\text{yoke}} = \frac{\Phi}{2}. \tag{2.65}$$

2.8.4 Step 2 : Select yoke flux density level

Choose a yoke flux density B_{yoke} consistent with the lamination and the design strategy. In steel-targeted design, B_{yoke} is typically selected in the range :

$$1.4 \text{ T} \leq B_{\text{yoke}} \leq 1.7 \text{ T},$$

and it is enforced to remain lower than the tooth flux density target (to avoid over-saturating the back iron relative to teeth).

2.8.5 Step 3 : Required yoke cross-sectional area

From the flux-density definition :

$$A_c = \frac{\Phi_{\text{yoke}}}{B_{\text{yoke}}}. \tag{2.66}$$

Here, A_c is the cross-sectional area of yoke iron needed to carry Φ_{yoke} at flux density B_{yoke}.

2.8.6 Step 4 : Yoke (core) radial thickness

Assuming the yoke cross-section is approximated by a rectangular area $A_c \approx d_{cs} L_i$, the required radial thickness is :

$$d_{cs} = \frac{A_c}{L_i}. \tag{2.67}$$

The effective iron length L_i is used (net stack length), not the gross mechanical stack.

2.8.7 Step 5 : Outer diameter from bore, slot height, and yoke thickness

The outer diameter is obtained by adding twice the slot height and twice the yoke thickness to the bore :

$$D_{\text{ext,calc}} = D + 2h_{\text{slot}} + 2d_{cs}. \tag{2.68}$$

Slot-height consistency (important). The method requires a consistent definition of h_{slot} from the chosen slot model :

- **Semi-open (or similar multi-step slot) :** $h_{\text{slot}} = h_{s0} + h_{s1} + h_{s2}$.
- **Open-slot pitch/area/ratio model :** $h_{\text{slot}} = h_s$ (directly solved by the open-slot geometry step).

This guarantees that the outer diameter reflects the same slot geometry used in the electromagnetic and copper-area phases.

2.9 Module 7 : Performance Estimation (Weights & Losses)

This final phase calculates the resistive parameters, component weights, and the resulting energy losses. These calculations rely heavily on the material properties of the selected conductors and electrical steel.

2.9.1 Algorithm Overview (Program Steps)

1. Read material properties (steel grade data and copper parameters).
2. Compute winding mean turn length, stator resistance, and copper losses (P_{js}).
3. Estimate copper mass from conductor area and total conductor length.
4. Estimate iron masses (teeth and yoke) using the chosen slot geometry.
5. Compute iron losses using specific-loss data and the estimated steel mass.

2.9.2 Symbols and Parameters

— ρ_{cu} : copper resistivity [Ω·m]
— ρ_{steel} : steel density [kg/m^3]
— L_{mt} : mean turn length [m]
— R_s : stator phase resistance [Ω]
— P_{js} : stator copper (Joule) losses [W]
— P_{kg} : specific iron loss [W/kg] at specified B and f

2.9.3 1. Reference Material Properties

The software utilizes a specific database of electrical steel grades to determine density (ρ_{steel}) and specific iron loss (P_{kg}).

TABLE 2.11 – Electrical steel grades : specific loss and density (used for iron-loss estimation)

Grade	**Specific Loss (P_{kg})** (W/kg @ 1.5T)	**Density** (kg/m^3)
M800-65A	8.00	7800
M600-50A	6.00	7750
M400-50A	4.00	7700
M270-35A	**2.70**	**7650**
M235-35A	2.35	7600

2.9.4 2. Winding Resistance and Copper Losses

Mean Turn Length (L_{mt})

The average length of a single winding turn includes the active length inside the slot and the overhangs (end-windings).

$$L_{mt} = 2 \cdot L + 2.3 \cdot \tau_{pole} + 0.24 \quad [m] \tag{2.69}$$

Stator Resistance (R_s)

Calculated at a reference temperature of 20°C using the resistivity of copper ($\rho_{cu} = 1.68 \times 10^{-8} \Omega \cdot m$).

$$R_s = \frac{\rho_{cu} \cdot L_{mt} \cdot T_{ph}}{A_s \times 10^{-6}} \quad [\Omega] \tag{2.70}$$

Where A_s is the conductor cross-sectional area [mm^2] (either $a_{c(std)}$ for wire or A_{strip} for strips).

Copper Losses (P_{js})

The Joule losses in the stator winding are calculated for all three phases :

$$P_{js} = 3 \cdot R_s \cdot I_s^2 \quad [W] \tag{2.71}$$

Copper Weight (W_{cus})

The total mass of the copper winding is estimated using the density of copper ($\approx 8900 kg/m^3$) :

$$W_{cus} = L_{mt} \cdot T_{ph} \cdot 3 \cdot A_s \cdot (8.9 \times 10^{-3}) \quad [kg] \tag{2.72}$$

2.9.5 3. Iron Weights and Losses

The iron losses are separated into Tooth losses and Core (Yoke) losses. The calculation of the tooth weight (W_t) depends on the slot geometry selected in Module 5.

Tooth Weight (W_t)

The total weight of the teeth is calculated as :

$$W_t = A_{tooth} \cdot L_i \cdot S_s \cdot \rho_{steel} \quad [kg] \tag{2.73}$$

The cross-sectional area A_{tooth} is derived differently for each slot type :

Method A : Semi-Open Slots (Wire) The tooth area consists of the main tooth body (A_{tb}) and the tooth shoe (A_{shoe}).

$$A_{tb} = h_{s2} \times w_{st} \tag{2.74}$$

$$A_{shoe} = w_{st0} \times h_{s0} + \left(\frac{w_{st0} + w_{st}}{2} \times h_{s1} \right) \tag{2.75}$$

$$A_{tooth} = A_{tb} + A_{shoe} \tag{2.76}$$

Where h_{s2} is the active slot depth, w_{st} is the tooth width, w_{st0} is the tooth tip width, and h_{s0}, h_{s1} are the opening and wedge heights.

Method B : Open Slots (Strip) For rectangular slots, the tooth is trapezoidal.

$$A_{tooth} = \frac{(w_{st1} + w_{st2}) \cdot h_s}{2} \tag{2.77}$$

Where w_{st1} is the tooth width at the bore, w_{st2} is the tooth width at the yoke, and h_s is the total slot height.

Core Weight (W_c)

The core (back iron) is modeled as a cylinder located at the mean diameter of the yoke.

$$D_{mean} = D + 2h_{total} + d_{cs} \quad [m] \tag{2.78}$$

$$W_c = \pi \cdot D_{mean} \cdot A_c \cdot \rho_{steel} \quad [kg] \tag{2.79}$$

Note : h_{total} is the total slot height (h_s for open slots ; $h_{s0} + h_{s1} + h_{s2}$ for semi-open slots).

Total Iron Losses (P_{iron})

Using the specific loss P_{kg} (W/kg) from Table 2.11 :

$$P_{it} = W_t \cdot P_{kg} \quad \text{(Tooth Loss)} \tag{2.80}$$

$$P_{ic} = W_c \cdot P_{kg} \quad \text{(Core Loss)} \tag{2.81}$$

$$P_{iron} = P_{it} + P_{ic} \tag{2.82}$$

Chapitre 3

Rotor Design Algorithmic Process

3.1 Introduction

3.1.1 Project Overview

This comprehensive report documents the development and implementation of a specialized software tool for the design and analysis of squirrel cage induction motor rotors. The project addresses the complex multidisciplinary challenge of rotor design, which involves the integration of electromagnetic theory, thermal considerations, mechanical constraints, and manufacturing practicalities.

The software tool implements a systematic methodology for determining optimal rotor parameters, including bar geometry, end-ring dimensions, material selection, and performance prediction. By automating the intricate calculations and constraint checking typically performed manually by motor design engineers, the tool significantly reduces design time while improving accuracy and reliability.

3.1.2 The Importance of Rotor Design in Induction Motors

Induction motors represent over 90% of industrial motor applications worldwide, with squirrel cage designs being predominant due to their robustness, reliability, and cost-effectiveness. The rotor constitutes the heart of these machines, with its design critically influencing :

- **Performance Characteristics** : Starting torque, breakdown torque, efficiency, and power factor
- **Operational Behavior** : Speed-torque characteristics, slip, and thermal performance
- **Manufacturing Considerations** : Castability of aluminum or copper, tooling requirements, and production yield
- **Lifecycle Costs** : Energy consumption, maintenance requirements, and reliability

3.1.3 Project Objectives

The primary objectives of this project are :

Technical Objectives

1. Develop a comprehensive mathematical model for rotor slot geometry optimization
2. Implement algorithms for automatic constraint checking and validation

3. Create a user-friendly interface for parameter input and result visualization
4. Integrate material databases with performance prediction capabilities
5. Enable rapid comparison of design alternatives

Practical Objectives

1. Reduce rotor design time from days to hours
2. Minimize design errors through automated verification
3. Provide educational value for engineering students and professionals
4. Support both new designs and redesign of existing motors
5. Generate comprehensive design documentation automatically

3.2 Rotor Bar Types in Three-Phase Squirrel-Cage Induction Motors

A three-phase squirrel-cage induction motor is widely used in industrial applications due to its robustness, simplicity, and low maintenance. The rotor of this motor consists of conductive bars embedded in the rotor core and short-circuited at both ends by end rings. The geometry of these rotor bars has a significant influence on the motor's electrical and mechanical performance.

3.2.1 Rectangular Rotor Bars

Rectangular rotor bars represent the simplest and most economical rotor design. These bars have a uniform cross-section and are typically shallow.

Characteristics :

— Simple manufacturing process
— Low starting torque
— High starting current
— Good efficiency at rated operating conditions

This type of rotor is commonly used in small motors and applications where high starting torque is not required.

3.2.2 Deep-Bar Rotor

In deep-bar rotors, the rotor bars are designed with a large depth compared to their width. This geometry increases the effect of current skin effect during startup.

At startup, the rotor current frequency is high, forcing the current to flow near the upper region of the bar, effectively increasing the rotor resistance. As the motor approaches synchronous speed, the rotor frequency decreases, allowing the current to penetrate deeper into the bar, thereby reducing resistance.

Advantages :

— Improved starting torque
— Reduced starting current
— Higher efficiency during steady-state operation

3.2.3 Double-Cage Rotor

The double-cage rotor consists of two separate rotor bars per slot : an outer high-resistance bar and an inner low-resistance bar.

Key benefits :

— Very high starting torque
— Low starting current
— Excellent efficiency at rated speed

This rotor design is particularly suitable for applications requiring heavy-load starting conditions, such as compressors and elevators.

3.2.4 Trapezoidal (inverted and direct) Rotor Bars

Trapezoidal rotor bars are a commonly used variation of the deep-bar rotor. In this design, the rotor bar has a trapezoidal cross-section, characterized by a narrower upper section near the air gap and a wider lower section embedded deeper in the rotor core.

Operating Principle

The trapezoidal geometry enhances the skin effect during startup. At high rotor frequencies, the current is concentrated in the narrow upper portion of the bar, increasing the effective rotor resistance. This results in higher starting torque and reduced inrush current.

During normal operation, the rotor slip becomes very small, and the rotor current frequency decreases. Consequently, the current distributes over the entire cross-sectional area of the bar, reducing the effective resistance and improving efficiency.

Trapezoidal and Inverted Trapezoidal Rotor Bars

Trapezoidal and inverted trapezoidal rotor bar geometries are both derived from tapered slot designs, but they are optimized for different operating requirements and power levels of induction motors. The main distinction between these two configurations lies in the distribution of the rotor bar cross-sectional area relative to the air gap, which directly influences the skin effect, effective rotor resistance, and torque production.

Trapezoidal Rotor Bars In the trapezoidal rotor bar configuration, the upper part of the bar near the air gap is narrower, while the lower part is wider. This geometry increases the effective rotor resistance at high slip values due to an enhanced skin effect during startup. As a result, the starting torque is increased while the starting current is limited.

This design is particularly well suited for small and medium-power induction motors, where moderate starting torque is sufficient and efficiency, smooth operation, and reduced mechanical stress are primary objectives. Additional advantages of the trapezoidal rotor bar include :

— Smooth torque–speed characteristic
— Improved thermal dissipation due to a wider bar base
— Enhanced mechanical strength and reduced vibration

Inverted Trapezoidal Rotor Bars In contrast, the inverted trapezoidal rotor bar features a wider upper section near the air gap and a narrower lower section. This configuration reduces the skin effect at startup by providing a larger current-carrying area close to the air gap, leading to a lower effective rotor resistance at high slip.

Consequently, inverted trapezoidal rotor bars are more suitable for high-power induction motors that require high electromagnetic torque and strong acceleration capabilities. The larger current interaction with the stator magnetic field results in higher torque production, making this design appropriate for heavy-duty industrial applications.

Design Considerations The choice between trapezoidal and inverted trapezoidal rotor bar geometries therefore depends on the intended motor application. Trapezoidal bars are optimized for induction motors requiring limited to moderate starting torque with smooth dynamic behavior, while inverted trapezoidal bars are designed for high-power motors where high torque capability is a dominant requirement. This distinction highlights the importance of rotor bar geometry in tailoring induction motor performance to specific operational needs.

3.2.5 Conclusion

The trapezoidal rotor bar offers an effective compromise between performance and manufacturing complexity. By exploiting the skin effect and varying the effective rotor resistance with slip, this design improves starting performance while maintaining high efficiency during steady-state operation. As a result, trapezoidal rotor bars are widely used in medium and large industrial three-phase squirrel-cage induction motors.

3.3 Magnetic Flux Density Distribution in Trapezoidal Rotor Bar Slots

The distribution of magnetic flux density in the rotor slots plays a critical role in determining the electromagnetic performance of a three-phase squirrel-cage induction motor. In particular, the geometry of the rotor bar strongly influences the concentration and path of magnetic flux within the rotor core.

3.3.1 Impact on Electromagnetic Performance

The flux density distribution in trapezoidal rotor bar slots has several beneficial effects :

— Reduction of magnetic saturation in deep rotor regions
— Improved torque production due to better flux coupling
— Lower iron losses and reduced local heating
— Enhanced smoothness of the electromagnetic torque

Furthermore, the concentration of flux near the air gap during startup enhances the interaction between stator and rotor magnetic fields, contributing to higher starting torque. At steady-state operation, the more uniform flux distribution improves efficiency and thermal performance.

3.3.2 Discussion

The trapezoidal rotor bar design provides an optimized compromise between flux concentration and magnetic saturation control. By shaping the rotor slot to guide the magnetic flux effectively, this design enhances both starting and steady-state performance of squirrel-cage induction motors. The flux density map confirms that trapezoidal slots allow controlled flux maxima while maintaining acceptable saturation levels within the rotor core.

3.3.3 Analysis of Starting Performance

Figures 21 and 22 illustrate the influence of rotor bar geometry on the starting performance of the induction motor. The electromagnetic torque versus slip characteristic and the speed evolution during startup are compared for different rotor slot configurations, including rectangular, trapezoidal, inverted trapezoidal, double-cage, and a reference case without skin effect.

3.3.4 Rationale for Choosing Trapezoidal Rotor Bars

The trapezoidal rotor bar represents a widely adopted industrial solution that offers an effective compromise between performance improvement and manufacturing complexity. Compared to conventional rectangular bars, the trapezoidal geometry enhances the skin

effect during startup by reducing the cross-sectional area near the air gap. This leads to an increase in the effective rotor resistance at high slip values, resulting in improved starting torque and reduced inrush current.

Additionally, the widening of the rotor bar toward the bottom of the slot promotes a more uniform magnetic flux distribution within the rotor core. This reduces local magnetic saturation and improves thermal dissipation, contributing to higher efficiency and better mechanical robustness. For these reasons, trapezoidal rotor bars are commonly used in medium- and high-power industrial induction motors, making them a relevant choice for this study.

3.3.5 Rationale for Choosing Inverted Trapezoidal Rotor Bars

The inverted trapezoidal rotor bar configuration was selected to provide a meaningful comparison and to highlight the influence of slot geometry on electromagnetic performance. In this design, the rotor bar has a wider upper section and a narrower lower section, leading to a different current and flux distribution compared to the standard trapezoidal geometry.

This inverted shape reduces the skin effect at startup by allowing a larger current-carrying area near the air gap. As a result, the effective rotor resistance at high slip is lower, which leads to reduced starting torque and longer acceleration time. Studying this configuration enables a clearer understanding of how geometric inversion affects torque-slip characteristics and dynamic response.

Moreover, the inverted trapezoidal model serves as a reference case to validate the impact of slot tapering direction on motor behavior. By comparing trapezoidal and inverted trapezoidal designs under identical operating conditions, the influence of rotor bar geometry can be isolated and analyzed in a systematic manner.

3.3.6 Overall Justification

The combined study of trapezoidal and inverted trapezoidal rotor bar geometries allows for a comprehensive evaluation of the role of slot shape in induction motor performance. While the trapezoidal design reflects an optimized and industrially relevant solution, the inverted trapezoidal configuration provides a contrasting case that emphasizes the importance of skin effect and flux concentration near the air gap. This comparative approach strengthens the validity of the conclusions drawn from the simulation results and supports informed rotor design decisions.

$\min_N r = max(10, int(Ns * 0.5)) max_N r = int(Ns * 1.5)$

for Nr in range($\min_N r, max_N r + 1$) : $if Nr invaleurs_{ae} viter : continuerapport = Ns/Nr if 0.6 <= rapport <= 1.4 : Considerthiscandidate$

3.3.7 Calculation Example

For a motor with $N_s = 36$ slots and $P = 2$ (4 poles) :

$$\text{Pole pairs} = 2$$
$$\text{Number of poles} = 4$$
$$\text{Recommended ratio} = 3/4 \text{ or } 4/3$$
$$N_{r1} = \frac{36}{3/4} = 48$$
$$N_{r2} = \frac{36}{4/3} = 27$$

Check constraints :
— $N_s - N_r$: $36 - 48 = -12$ (not in prohibited list)
— $N_s - N_r$: $36 - 27 = 9$ (not in prohibited list)
— Neither 27 nor 48 equals $N_s \pm 2P = 36 \pm 4 = \{32, 40\}$
Both 27 and 48 are valid candidates.

3.3.8 Algorithm Flowchart

Process :

1. Input N_s and P
2. Calculate prohibited values
3. Generate candidate N_r values
4. Filter by ratio constraints
5. Filter by prohibited values
6. Sort candidates by $|N_s - N_r|$
7. Present top recommendations

3.3.9 Implementation in Main Program

Listing 3.1 – Main program integration

```
# In calcul_dimensions_rotor()
print("\nChoix m thode Nr:")
print("  1) Classique")
print("  2) Harmonique")

if c == 1:
    rec, _ = selection_nr_classique(Ns, P)
    methode = "Classique"
elif c == 2:
    rec, _ = selection_nr_harmonique(Ns, P)
    methode = "Harmonique"

if rec:
    choix = int(input("Choisir Nr index (1..10) ou 0 manuel: "))
    if choix == 0:
        Nr = int(input("Nr (barres rotor): "))
    else:
        Nr = rec[max(1, min(choix, len(rec[:10]))) - 1][0]
```

3.3.10 Practical Guidelines

1. **Starting point** : Use $N_r \approx 0.7 - 1.3 \times N_s$
2. **Avoid** : $N_r = N_s$, $N_r = N_s \pm 1$, $N_r = N_s \pm 2P$
3. **Prefer** : $N_r < N_s$ for lower harmonic content
4. **Check** : $|N_s - N_r|$ should not be too small (3 recommended)
5. **Verify** : No integer multiple relationships
6. **Consider** : Skewing can mitigate some harmonic issues

3.3.11 Common Pitfalls to Avoid

Condition	Problem Caused
$N_r = N_s$	Severe synchronous locking, motor won't start
$N_r = N_s \pm 2P$	Strong slot harmonics, vibration
$N_r = N_s \pm 1$	Beat frequency vibrations
$\|N_s - N_r\|$ too small	Increased harmonic content
Integer ratios	Subharmonic vibrations

TABLE 3.1 – Problematic slot number combinations

3.3.12 Advanced Considerations

Effect on Starting Torque

$$\boxed{T_{start} \propto \frac{\sin\left(\frac{\pi P}{N_r}\right)}{\frac{\pi P}{N_r}}} \tag{3.1}$$

Purpose : Shows how N_r affects the starting torque through the winding factor.

Harmonic Order Calculation

$$\boxed{\nu = \frac{kN_s \pm P}{P}} \tag{3.2}$$

where ν is the harmonic order and k is an integer.

Optimal Range

For best performance :

$$\boxed{0.8 \leq \frac{N_r}{N_s} \leq 0.9 \quad \text{or} \quad 1.1 \leq \frac{N_r}{N_s} \leq 1.2} \tag{3.3}$$

This range typically provides good torque characteristics while minimizing harmonics.

Listing 3.2 – Advanced validation in code

```
# Additional validation could include:
def validate_nr_choice(Ns, Nr, P):
```

```
poles = 2 * P
ratio = Ns / Nr

# Check optimal range
if not (0.8 <= ratio <= 0.9 or 1.1 <= ratio <= 1.2):
    print("Warning: Ratio outside optimal range")

# Calculate harmonic orders
for k in range(1, 5):
    harmonic = (k * Ns - P) / P
    if abs(harmonic - round(harmonic)) < 0.1:
        print(f"Warning: Strong harmonic at order {harmonic:.1f}")
```

3.4 Geometrical Description of the Trapezoidal Rotor Bar

The geometrical configuration of the rotor bar plays a fundamental role in determining the electromagnetic behavior of squirrel-cage induction machines. In particular, the shape and dimensions of the rotor slot directly influence the distribution of magnetic flux density, current density, leakage flux, and electromagnetic torque.

In this study, a trapezoidal rotor bar geometry is adopted to represent a realistic industrial rotor design. The trapezoidal shape is defined by a narrow upper width near the rotor surface and a wider lower width at the base of the slot, combined with a specified slot height. This gradual variation in cross-sectional area allows improved mechanical strength of the rotor teeth while enabling better control of electromagnetic flux paths within the rotor core.

The main geometrical parameters of the trapezoidal rotor slot include the upper slot width, lower slot width, slot height, slot opening, and the effective bar cross-sectional area. These parameters are selected to ensure an optimal compromise between magnetic performance, thermal dissipation, and manufacturability. A wider slot base enhances current distribution and heat transfer, while the reduced upper opening limits air-gap harmonic content and local magnetic saturation.

Accurate definition of the rotor bar geometry is essential for reliable magnetic field analysis. The trapezoidal profile provides smoother magnetic flux transitions between the rotor teeth and the air gap, reducing flux crowding and improving the uniformity of magnetic induction. Consequently, this geometry is particularly well suited for finite element analysis and analytical modeling of induction machines.

The magnetic flux density distribution resulting from the selected trapezoidal rotor bar geometry is illustrated in Figure 3.1, where both the spatial flux map and the corresponding color scale are presented.

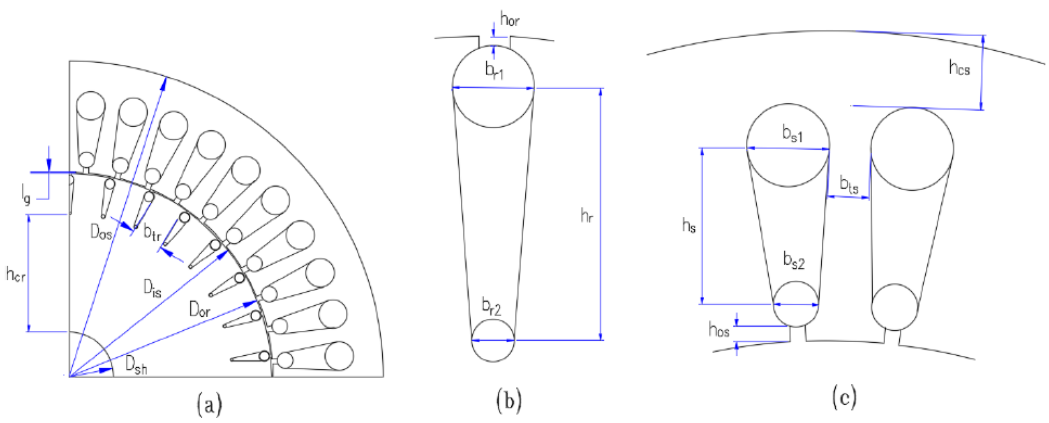

Figure 1: Machine section with (a) stator and rotor, (b) rotor slot, and (c) stator slot.

FIGURE 3.1 – Electromagnetic torque versus slip for different rotor bar geometries

3.5 Geometric Constraints and Parameters

3.5.1 Constraint Constants Definition

Parameter	Description	Value	Unit
EPS_R	Minimum radius difference	0.01	mm
GAP_D	Minimum distance margin	0.10	mm
D_MAX_MARGIN	Maximum diameter margin	0.20	mm
MAX_R2_R1_RATIO	Maximum r2/r1 ratio	0.85	–
MIN_R2_R1_RATIO	Minimum r2/r1 ratio	0.30	–
LSTACK_D_RATIO	Stack length to diameter ratio	0.53	–

TABLE 3.2 – Fixed geometric constraint parameters

3.5.2 Radius Ratio Constraints

$$\boxed{r_1 > r_2 + \text{EPS_R}} \tag{3.4}$$

Purpose : Ensures the upper radius is always larger than the lower radius with a safety margin to prevent geometric interference.

$$\boxed{R_{\min} \leq \frac{r_2}{r_1} \leq R_{\max}} \tag{3.5}$$

Purpose : Maintains reasonable proportions between radii for manufacturability and optimal current distribution.

Listing 3.3 – Constraint checking in code

```
# Constraint verification in solve_slot_geometry_simple()
if not (r1 > r2 + EPS_R):
    continue  # Skip this combination
if not (MIN_R2_R1_RATIO <= ratio <= MAX_R2_R1_RATIO):
    continue  # Skip this combination
```

3.5.3 Distance Constraints

$$\boxed{d_{rb} \geq r_1 + r_2 + \text{GAP_D}} \tag{3.6}$$

$$\boxed{d_{rb} \leq \frac{D}{2} - g - h_{r0} - \text{D_MAX_MARGIN}} \tag{3.7}$$

Purpose : Ensures the slot fits within the rotor diameter while maintaining adequate clearance.

Listing 3.4 – Distance constraint calculation

```
# Calculation of maximum allowed distance
d_max = (D/2) - g - h_r0 - D_MAX_MARGIN

# Verification
```

```
d_min = r1 + r2 + GAP_D
if d_min > d_max:
    continue  # This combination is impossible
```

3.6 Slot Geometry Equations

3.6.1 Angle Calculation

$$\boxed{\alpha = \frac{2\pi}{N_r}} \tag{3.8}$$

Purpose : Calculates the angular opening of each slot based on the number of rotor bars.

Listing 3.5 – Angle calculation in code

```
# In solve_slot_geometry_simple()
alpha = 2 * math.pi / Nr
alpha_deg = alpha * 180 / math.pi
cos_half = math.cos(alpha/2)
```

3.6.2 Geometric Area Calculation

The total slot area A_{geo} is composed of four parts :

$$\boxed{A_1 = \frac{1}{2} r_1^2 \theta_1} \tag{3.9}$$

where $\theta_1 = \frac{\pi}{2} + \frac{\alpha}{2}$

$$\boxed{A_2 = \frac{1}{2} r_2^2 \theta_2} \tag{3.10}$$

where $\theta_2 = \frac{\pi}{2} - \frac{\alpha}{2}$

$$\boxed{A_3 = h_{r0} \times b_{r0}} \tag{3.11}$$

$$\boxed{A_4 = (r_1 + r_2) \times (d_{rb} - r_1 - r_2) \times \cos\left(\frac{\alpha}{2}\right)} \tag{3.12}$$

$$\boxed{A_{geo} = A_1 + A_2 + A_3 + A_4} \tag{3.13}$$

Listing 3.6 – Area calculation function

```
def compute_a_Arb_detailed(r1, r2, alpha, h_r0, b_r0, d_rb):
    """Calculates the geometric section with all terms"""
    angle_r1 = math.pi/2 + alpha/2
    angle_r2 = math.pi/2 - alpha/2
    cos_half = math.cos(alpha/2)

    # Circular sectors
    A1 = 0.5 * r1**2 * angle_r1
```

```
A2 = 0.5 * r2**2 * angle_r2

# Central rectangle
A3 = h_r0 * b_r0

# Trapezoid
A4 = (r1 + r2) * (d_rb - r1 - r2) * cos_half

total = A1 + A2 + A3 + A4

return total, {
    "secteur_r1": A1,
    "secteur_r2": A2,
    "rectangle": A3,
    "trapeze": A4,
    "total": total
}
```

3.7 Electrical Calculations

3.7.1 Rotor Current Calculation

$$\boxed{I_r = 0.85 \times I_{ph}} \tag{3.14}$$

Purpose : Approximates rotor current as 85% of stator phase current based on typical induction motor characteristics.

Listing 3.7 – Rotor current calculation

```
# In calcul_dimensions_rotor()
Ir = 0.85 * Iph
show_calc(" tape  E1) Courant rotorique Ir (hypoth se)",
          "Ir = 0.85 Iph ",
          f"Ir = 0.85  {Iph}",
          f"Ir = {Ir:.3f} A")
```

3.7.2 Bar Current Calculation

$$\boxed{I_b = \frac{I_r \times K_{ws} \times N_s \times Z_s}{2 \times k_{wr} \times N_r \times Z_r}} \tag{3.15}$$

Where :

- K_{ws} : Stator winding factor (typically 0.955)
- k_{wr} : Rotor winding factor (typically 1.0 for cage)
- Z_s : Conductors per stator slot
- Z_r : Conductors per rotor slot (typically 1)

Listing 3.8 – Bar current calculation

```
# In calcul_dimensions_rotor()
Ib = Ir * Kws * Ns * Zs / (2 * kwr * Nr * Zr)
show_calc(" tape  E2) Courant de barre Ib",
          "Ib = Ir Kws Ns Zs / (2   kwrNrZr   )",
          f"Ib = {Ir:.3f}  {Kws}  {Ns}  {Zs}/(2  {kwr}  {Nr}  {Zr})",
          f"Ib = {Ib:.3f} A")
```

3.7.3 Required Bar Cross-Section

$$\boxed{A_{b,req} = \frac{I_b}{J_b}} \tag{3.16}$$

Purpose : Calculates the minimum copper cross-section needed based on current density J_b.

Listing 3.9 – Required cross-section calculation

```
# In calcul_dimensions_rotor()
Ab_req = Ib / Jb
show_calc(" tape  E3) Section   lectrique   Ab_elec",
          "Ab_elec = Ib/Jb",
```

```
            f"Ab_elec = {Ib:.3f}/{Jb}",
            f"Ab_elec = {Ab_req:.3f} mm ")
```

3.8 Magnetic Circuit Equations

3.8.1 Maximum Rotor Tooth Flux

$$\boxed{\phi_{rt,\max} = \phi \times \sin\left(\frac{\pi \times P}{N_r}\right)} \tag{3.17}$$

Purpose : Calculates the maximum flux in rotor teeth considering spatial distribution.

Listing 3.10 – Maximum tooth flux calculation

```
# In calcul_dimensions_rotor()
fluxrt_max = phi * math.sin(math.pi * P / Nr)
show_calc(" tape M1) Flux max dent rotor (corrig )",
            " _rt_max = sin ( P /Nr)",
            f" _rt_max = {phi} sin ( {P}/{Nr})",
            f" _rt_max = {fluxrt_max:.6f} Wb")
```

3.8.2 Rotor Tooth Width

$$\boxed{W_{rt} = 1000 \times \frac{\phi_{rt,\max}}{B_{rt} \times L_m \times K_i}} \tag{3.18}$$

Where :

- $L_m = L/1000$: Stack length in meters
- K_i : Iron stacking factor (typically 0.95)
- B_{rt} : Flux density in rotor teeth

Listing 3.11 – Rotor tooth width calculation

```
# In calcul_dimensions_rotor()
L_m = L / 1000.0
Wrt = 1000.0 * (fluxrt_max / (Brt * L_m * Ki))
show_calc(" tape M2) Largeur dent rotor Wrt",
            "Wrt = [ _rt_max /(Brt L Ki)] 1000 ",
            f"Wrt = [{fluxrt_max:.6f}/({Brt} {L_m:.3f} {Ki})] 1000 ",
            f"Wrt = {Wrt:.3f} mm")
```

3.8.3 Rotor Yoke Width

$$\boxed{W_{ry} = 1000 \times \frac{\phi/2}{B_{ry} \times L_m \times K_i}} \tag{3.19}$$

Purpose : Calculates the width needed in the rotor yoke to carry half the pole flux.

Listing 3.12 – Rotor yoke width calculation

```
# In calcul_dimensions_rotor()
Wry = 1000.0 * ((phi/2) / (Bry * L_m * Ki))
show_calc(" tape  M3) Largeur culasse rotor Wry",
          "Wry = [(   /2)/(Bry L Ki)] 1000 ",
          f"Wry = [({phi}/2)/({Bry}  {L_m:.3f}  {Ki})] 1000 ",
          f"Wry = {Wry:.3f} mm")
```

3.9 End Ring Calculations

3.9.1 End Ring Current

$$\boxed{I_e = \frac{N_r \times I_b}{\pi \times P}} \tag{3.20}$$

Purpose : Calculates the current flowing in each end ring segment.

Listing 3.13 – End ring current calculation

```
# In calculer_anneau_court_circuit()
Ie = (Nr * Ib) / (math.pi * P)
show_calc(" tape  A1) Courant d'anneau Ie",
          "Ie = (Nr    Ib) / (      P)",
          f"Ie = ({Nr}    {Ib:.3f}) / (      {P})",
          f"Ie = {Ie:.3f} A")
```

3.9.2 Required End Ring Cross-Section

$$\boxed{A_{e,req} = \frac{I_e}{J_e}} \tag{3.21}$$

Where : J_e is the current density in end rings (typically higher than in bars).

Listing 3.14 – End ring cross-section calculation

```
# In calculer_anneau_court_circuit()
aAeb_req = Ie / Je
show_calc(" tape  A2) Section n cessaire aAeb",
          "aAeb = Ie / Je",
          f"aAeb = {Ie:.3f} / {Je:.3f}",
          f"aAeb = {aAeb_req:.3f} mm ")
```

3.9.3 Actual End Ring Cross-Section

$$\boxed{A_{e,actual} = h_{er} \times b_{er}} \tag{3.22}$$

Purpose : Calculates the actual cross-sectional area based on chosen dimensions.

Listing 3.15 – Actual end ring cross-section calculation

```
# In calculer_anneau_court_circuit()
aAeb_reel = h_er * b_er
Je_reel = Ie / aAeb_reel if aAeb_reel else float("inf")
show_calc(" tape  A4) V rification section r elle",
          "aAeb_reel = h_er    b_er",
          f"aAeb_reel = {h_er}    {b_er}",
          f"aAeb_reel = {aAeb_reel:.3f} mm ")
```

3.10 Resistance Calculations

3.10.1 Bar Length with Skew

$$\boxed{L_{bar} = \frac{L_{stack}}{\cos(\theta_{skew})}} \tag{3.23}$$

Purpose : Calculates the actual bar length considering skew angle.

Listing 3.16 – Bar length with skew calculation

```
# In calculer_resistance_barre()
if skew_angle != 0:
    skew_rad = math.radians(skew_angle)
    Lbar = Lstack / math.cos(skew_rad)
    show_calc(" tape  B1) Longueur effective de barre (avec inclinaison)",
              "Lbar = Lstack / cos( _skew )",
              f"Lbar = {Lstack:.3f} / cos({skew_angle}  )",
              f"Lbar = {Lbar:.3f} mm")
else:
    Lbar = Lstack
```

3.10.2 Bar Resistance

$$\boxed{R_{bar} = \rho \times \frac{L_{bar}}{A_{bar}}} \tag{3.24}$$

Where :

- ρ : Resistivity of bar material (·m)
- L_{bar} : Bar length (m)
- A_{bar} : Bar cross-section (m^2)

Note : In code, we convert ρ from ·m to ·mm^2/m using $\rho_{mm} = \rho \times 10^6$

Listing 3.17 – Bar resistance calculation

```
# In calculer_resistance_barre()
rho_mm_bar = rho_m_bar * 1e6  #    mm    /m
Rbar = rho_mm_bar * (Lbar / 1000) / Arb
show_calc(" tape  B3) R sistance d'une barre Rbar",
          "Rbar = (       10    )     (Lbar/1000) / Arb",
          f"Rbar = ({rho_m_bar:.2e}     10    )     ({Lbar:.3f}/1000) / {Arb:.
          f"Rbar = {Rbar:.6e}    ")
```

3.10.3 Total Parallel Bar Resistance

$$\boxed{R_{bar,total} = \frac{R_{bar}}{N_r}} \tag{3.25}$$

Purpose : Calculates the equivalent resistance of all bars in parallel.

Listing 3.18 – Total bar resistance calculation

```
# In calculer_resistance_barre()
Rbar_total_parallel = Rbar / Nr if Nr > 0 else float('inf')
show_calc(" tape  B5) R sistance totale des barres (en parall le)",
          "Rbar_total_parallel = Rbar / Nr",
          f"Rbar_total_parallel = {Rbar:.6e} / {Nr}",
          f"Rbar_total_parallel = {Rbar_total_parallel:.6e}    ")
```

3.10.4 End Ring Resistance

Mean Diameter

$$\boxed{D_{er} = D_r + 2 \times L_g + b_{er}} \tag{3.26}$$

Where :
- D_r : Rotor diameter
- L_g : Air gap length
- b_{er} : End ring width

Listing 3.19 – End ring mean diameter calculation

```
# In calculer_resistance_anneau()
Der = Dr + 2 * Lg + b_er
show_calc(" tape  R1) Diam tre moyen de l'anneau Der",
          "Der = Dr + 2 Lg  + b_er",
          f"Der = {Dr:.3f} + 2  {Lg:.3f} + {b_er}",
          f"Der = {Der:.3f} mm")
```

End Ring Length

$$\boxed{L_{er} = \pi \times D_{er}} \tag{3.27}$$

Listing 3.20 – End ring length calculation

```
# In calculer_resistance_anneau()
Ler = math.pi * Der
show_calc(" tape  R2) Longueur de l'anneau Ler",
          "Ler =        Der",
          f"Ler =        {Der:.3f}",
          f"Ler = {Ler:.3f} mm")
```

End Ring Resistance

$$\boxed{R_{er} = \rho \times \frac{L_{er}}{A_{er}}} \tag{3.28}$$

Where : A_{er} is the end ring cross-sectional area.

Listing 3.21 – End ring resistance calculation

```
# In calculer_resistance_anneau()
rho_mm = rho_m * 1e6  #    mm    /m
Rer = rho_mm * (Ler / 1000) / Aer
show_calc(" tape  R4) R sistance de l'anneau Rer",
          "Rer = (       10    )     (Ler/1000) / Aer",
          f"Rer = ({rho_m:.2e}     10    )     ({Ler:.3f}/1000) / {Aer:.3f}",
          f"Rer = {Rer:.6e}    ")
```

3.11 Loss Calculations

3.11.1 Bar Joule Losses

$$\boxed{P_{bar} = N_r \times I_b^2 \times R_{bar}} \tag{3.29}$$

Purpose : Calculates total resistive losses in all rotor bars.

Listing 3.22 – Bar losses calculation

```
# In calculer_resistance_barre()
P_bar = Nr * (Ib ** 2) * Rbar
show_calc(" tape  B6) Pertes Joule dans les barres",
          "P_bar = Nr     Ib      Rbar",
          f"P_bar = {Nr}     ({Ib:.3f})       {Rbar:.6e}",
          f"P_bar = {P_bar:.3f} W")
```

3.11.2 End Ring Joule Losses

$$\boxed{P_{er} = 2 \times I_e^2 \times R_{er}} \tag{3.30}$$

Purpose : Calculates resistive losses in both end rings (two rings in parallel).

Listing 3.23 – End ring losses calculation

```
# In calcul_dimensions_rotor()
P_er = 2 * (Ie ** 2) * Rer
show_calc("Pertes Joule dans les anneaux",
          "P_er = 2     Ie      Rer",
          f"P_er = 2     ({Ie:.3f})       {Rer:.6e}",
          f"P_er = {P_er:.3f} W")
```

3.11.3 Total Rotor Losses

$$\boxed{P_{rotor,total} = P_{bar} + P_{er}} \tag{3.31}$$

Listing 3.24 – Total rotor losses calculation

```
# In calcul_dimensions_rotor()
P_rotor_total = P_bar + P_er
show_calc("Pertes totales rotoriques",
          "P_rotor_total = P_bar + P_er",
          f"P_rotor_total = {P_bar:.3f} + {P_er:.3f}",
          f"P_rotor_total = {P_rotor_total:.3f} W")
```

3.12 Mechanical Dimensions

3.12.1 Air Gap Calculation

$$\boxed{L_g = 0.2 + 2 \times \sqrt{\frac{D \times L}{1,000,000}}} \tag{3.32}$$

Purpose : Empirical formula for air gap length based on machine dimensions.

Listing 3.25 – Air gap calculation

```
# In calcul_dimensions_rotor()
Lg = 0.2 + 2 * math.sqrt(D * L / 1_000_000)
g = Lg
show_calc(" tape  G1) Entrefer Lg",
          "Lg = 0.2 + 2      (D L/1e6)",
          f"Lg = 0.2 + 2      ({D}  {L}/1e6)",
          f"Lg = {Lg:.3f} mm")
```

3.12.2 Rotor Diameter

$$\boxed{D_r = D - 2 \times L_g} \tag{3.33}$$

Listing 3.26 – Rotor diameter calculation

```
# In calcul_dimensions_rotor()
Dr = D - 2 * Lg
show_calc(" tape  G2) Diam tre rotor Dr",
          "Dr = D - 2 Lg ",
          f"Dr = {D} - 2  {Lg:.3f}",
          f"Dr = {Dr:.3f} mm")
```

3.12.3 Rotor Slot Pitch

$$\boxed{sp_2 = \frac{\pi \times D_r}{N_r}} \tag{3.34}$$

Purpose : Calculates the circumferential distance between rotor slots.

Listing 3.27 – Rotor slot pitch calculation

```
# In calcul_dimensions_rotor()
sp2 = math.pi * Dr / Nr
show_calc(" tape  G3) Pas encoches rotor sp2",
          "sp2 =   Dr  /Nr",
          f"sp2 =      {Dr:.3f}/{Nr}",
          f"sp2 = {sp2:.3f} mm")
```

3.12.4 Stack Length Determination

Three methods available :

Method 2 : Direct Input

$$\boxed{L_{stack} = \text{Stator data}} \tag{3.35}$$

Method 3 : Equal to Total Length

$$\boxed{L_{stack} = L} \tag{3.36}$$

Listing 3.28 – Stack length determination

```
# In calcul_dimensions_rotor()
if methode_Lstack == 1:
    # Method 1: Constant ratio
    Lstack = LSTACK_D_RATIO * D
elif methode_Lstack == 2:
    # Method 2: Direct input
    Lstack = float(input("Lstack (stack length, mm): "))
elif methode_Lstack == 3:
    # Method 3: Equal to total length
    Lstack = L
```

3.13 Search Algorithm Equations

3.13.1 Search Range for r_1

$$\boxed{r_{1,\min} = \max(0.5, 0.15 \times W_{rt})} \tag{3.37}$$

$$\boxed{r_{1,\max} = \min(0.45 \times W_{rt}, 0.3 \times d_{\max})} \tag{3.38}$$

Where : $d_{\max} = \frac{D}{2} - g - h_{r0} - \text{D_MAX_MARGIN}$

Listing 3.29 – R1 search range calculation

```
# In solve_slot_geometry_simple()
r1_min = max(0.5, Wrt * 0.15)
r1_max = min(Wrt * 0.45, d_max * 0.3)
r1_max = max(r1_min + 0.5, r1_max)  # Ensure minimum range
```

3.13.2 Error Calculation

$$\boxed{\epsilon = \frac{|A_{geo} - A_{target}|}{A_{target}} \times 100\%} \tag{3.39}$$

Purpose : Calculates percentage error between geometric and target areas.

Listing 3.30 – Error calculation

```
# In solve_slot_geometry_simple()
error = abs(Ab_calc - Ab_target) / Ab_target * 100
if error < best_error:
    best_error = error
    best_solution = (r1, r2, d_test, Ab_calc, parts)
```

3.13.3 Iterative Search for d_{rb}

Algorithm increments d_{rb} until target area is reached : [1] $d_{test} \leftarrow r_1 + r_2 + \text{GAP_D}$ $i = 1$ to 50 $A_{calc} \leftarrow \text{compute_area}(r_1, r_2, d_{test})$ $A_{calc} \geq A_{target}$ d_{test}, A_{calc} $d_{test} \leftarrow d_{test} + 0.5$ $d_{test} > d_{\max}$ **break**

Listing 3.31 – Iterative search implementation

```
# In solve_slot_geometry_simple()
d_test = d_min
max_iter = 50
found = False

for iter in range(max_iter):
    Ab_calc, parts = compute_a_Arb_detailed(r1, r2, alpha, h_r0, b_r0, d_te

    if Ab_calc >= Ab_target:
        # Solution found
        error = abs(Ab_calc - Ab_target) / Ab_target * 100
        # Store best solution
        break
    else:
        # Increase d_rb to increase area
        d_test += 0.5
        if d_test > d_max:
            break
```

3.14 Material Resistivity Data

Material	Resistivity ρ ($\cdot$m)	Conductivity
Electrolytic copper	1.68×10^{-8}	Very high
Pure aluminum	2.82×10^{-8}	High
Cast aluminum (Al-Si)	3.4×10^{-8}	Medium
Brass (Cu-Zn)	7.0×10^{-8}	Low
Bronze (Cu-Sn)	8.0×10^{-8}	Low

TABLE 3.3 – Material resistivity table used in the program

Resistivity Conversion :

$$\boxed{\rho_{mm} = \rho \times 10^6 \quad [\Omega \cdot mm^2/m]} \tag{3.40}$$

Purpose : Converts resistivity from $\cdot$m to $\cdot$mm^2/m for easier calculation with mm dimensions.

Listing 3.32 – Resistivity table in code

```
RESISTIVITY_TABLE = {
    "Electrolytic copper": 1.68e-8,
    "Pure aluminum": 2.82e-8,
    "Cast aluminum (Al-Si)": 3.4e-8,
    "Brass (Cu-Zn)": 7.0e-8,
    "Bronze (Cu-Sn)": 8.0e-8
}

# Conversion to mm/m
rho_mm = rho_m * 1e6
```

3.15 Complete Solution Algorithm

Complete Rotor Slot Geometry Solution [1] $A_{target}, N_r, D, g, h_{r0}, b_{r0}, W_{rt}$ Optimal (r_1, r_2, d_{rb}) or None $\alpha \leftarrow 2\pi/N_r$ $d_{\max} \leftarrow D/2 - g - h_{r0} - \Delta_{\max}$ Calculate $r_{1,\min}$ and $r_{1,\max}$ using Equations 3.37, 3.38 Generate r_1 candidates from $[r_{1,\min}, r_{1,\max}]$ $R_{candidates} \leftarrow \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8\}$ $best_solution \leftarrow$ None, $best_error \leftarrow \infty$ each r_1 in r_1 candidates each $ratio$ in $R_{candidates}$ $r_2 \leftarrow r_1 \times ratio$ not $(r_1 > r_2 + \epsilon_R)$ **continue** not $(R_{\min} \leq ratio \leq R_{\max})$ **continue** $d_{\min} \leftarrow r_1 + r_2 + \delta_D$ $d_{\min} > d_{\max}$ **continue** $d_{test} \leftarrow d_{\min}$ $i = 1$ to 50 $A_{calc} \leftarrow$ compute_area(r_1, r_2, d_{test}) $A_{calc} \geq A_{target}$ $\epsilon \leftarrow |A_{calc} - A_{target}|/A_{target} \times 100\%$ $\epsilon < best_error$ $best_error \leftarrow \epsilon$, $best_solution \leftarrow (r_1, r_2, d_{test}, A_{calc})$ $\epsilon \leq tolerance$ $best_solution$ **break** $d_{test} \leftarrow d_{test} + 0.5$ $d_{test} > d_{\max}$ **break** $best_solution$ is None Try relaxed constraints still no solution None $best_solution$

3.16 Summary of Key Parameters

Parameter	Description	Typical Value	Unit
N_s	Number of stator slots	stator data input	–
N_r	Number of rotor bars	Calculated/Input	–
P	Pole pairs	User input	–
D	Stator inner diameter	stator data input	mm
L	Total rotor length	stator data input	mm
L_{stack}	Stack length	Calculated	mm
I_{ph}	Phase current	stator data input	A
Z_s	Conductors per stator slot	stator data input	–
ϕ	Flux per pole	stator data input	Wb
B_{rt}	Flux density in rotor teeth	stator data input	T
B_{ry}	Flux density in rotor yoke	stator data input	T
K_{ws}	Stator winding factor	0.955	–
k_{wr}	Rotor winding factor	1.0	–
Z_r	Conductors per rotor slot	1	–
K_i	Iron stacking factor	0.95	–
J_b	Bar current density	3.0-5.0	A/mm^2
J_e	End ring current density	5.0-7.0	A/mm^2
h_{r0}	Central rectangle height	0.5	mm
b_{r0}	Central rectangle width	0.8	mm
ρ	Material resistivity	See Table 3.3	·m

TABLE 3.4 – Summary of all parameters used in the calculations

Chapitre 4

Efficiency Analysis Tab : MMF, No-Load Current, Losses, and Actual Efficiency

4.1 What the User Sees in the Tab

The interface is centered around the **magnetizing requirement** of the machine :

- Ampere-turns (MMF) in :
  - stator core : **ATSC**
  - stator teeth : **ATST**
  - rotor core : **ATRC**
  - rotor teeth : **ATRT**
  - air gap (Carter-corrected) : **ATg**
- Total MMF : **AT_total**
- Magnetizing current : **Im**
- No-load losses and no-load current : **Pnl**, **Iw**, **I0**, **Pf0**
- Actual efficiency from total losses : **efficiency**
- Nominal slip estimated from losses : **g**

The tab only appears when **both stator and rotor designs are completed** (the program checks that both result structures exist).

4.2 Algorithm Overview (Program Steps)

1. Read stator design objects (specifications, geometry, winding, validation info) and rotor results.
2. Select the yoke flux density target B_{yoke} and interpolate $H(B_{yoke})$ from the selected steel B–H curve.
3. Compute **stator core MMF ATSC**.
4. Compute **stator tooth MMF ATST** using a depth-varying tooth flux profile and segment integration.
5. Compute **rotor core MMF ATRC** using rotor mean diameter.
6. Compute **rotor tooth MMF ATRT** using a 3-segment rotor-tooth model and a depth-varying tooth flux profile.
7. Compute **airgap MMF ATg** using Carter-corrected effective airgap length and the airgap flux density.

8. Sum all MMF terms to get **AT_total**.
9. Compute **magnetizing current Im** from **AT_total**, poles P, winding factor k_w and turns per phase T_{ph}.
10. Collect losses (stator copper, stator iron, rotor copper, friction) and also compute rotor iron losses inside the tab.
11. Compute no-load power **Pnl**, then compute no-load current **I0** and no-load power factor **Pf0**.
12. Compute **Actual efficiency** from rated power and total losses.
13. Estimate **Nominal slip g** from rotor copper losses and transmitted power.

4.3 Step 1 : Steel B–H curve and the value of H

The program chooses a yoke flux density B_{yoke} (typically taken from the forward design targets ; fallback used if missing) and interpolates the corresponding magnetic field intensity H :

$$H = H(B_{yoke})$$

This H is used for both stator and rotor **core** MMF calculations in this tab.

4.4 B–H Tables Used (B in T, H in A/m)

4.4.1 Grade M800-65A

Table 4.1 – B–H curve for M800-65A

B (T)	0.10	0.50	1.00	1.20	1.40	1.50	1.60	1.65	1.70	1.75
H (A/m)	35	65	115	170	290	400	600	850	1500	3000

4.4.2 Grade M600-50A

Table 4.2 – B–H curve for M600-50A

B (T)	0.10	0.50	1.00	1.20	1.40	1.50	1.60	1.65	1.70	1.75	1.80
H (A/m)	30	55	95	130	210	280	400	550	900	1800	4000

4.4.3 Grade M400-50A

Table 4.3 – B–H curve for M400-50A

B (T)	0.10	0.50	1.00	1.20	1.40	1.50	1.60	1.65	1.70	1.75	1.80
H (A/m)	25	45	80	105	160	210	300	400	650	1200	2500

4.4.4 Grade M270-35A

Table 4.4 – B–H curve for M270-35A

B (T)	0.10	0.50	1.00	1.20	1.40	1.50	1.60	1.65	1.70	1.75	1.80
1.85											
H (A/m)	20	35	65	85	120	150	210	270	400	650	1200
2500											

4.4.5 Grade M235-35A

Table 4.5 – B–H curve for M235-35A

B (T)	0.10	0.50	1.00	1.20	1.40	1.50	1.60	1.65	1.70	1.75	1.80	1.85
1.90												
H (A/m)	18	30	55	70	95	120	160	200	280	420	700	1300
3000												

4.5 Step 2 : Stator Core Ampere-Turns ATSC

The tab uses :

$$\boxed{ATSC = \frac{\pi\, D_{mean}\, H}{P \times 3}} \tag{4.1}$$

where :

- P is the **number of poles** (not pole pairs),
- D_{mean} is a **mean stator diameter** (in meters) reused from the core-weight computation when available,
- division by 3 is applied exactly as in the program (per-phase sharing).

4.6 Step 3 : Stator Tooth Ampere-Turns ATST

The stator tooth MMF is computed from a **depth-varying tooth flux profile**. The tab explicitly uses :

$$\boxed{\phi(z) = \phi_{max}\left(1 - \frac{z}{h_t + 0.7\, d_{cs}}\right)} \tag{4.2}$$

with :

- ϕ_{max} : maximum tooth flux used by the stator slot/tooth module (the tab reads it from validation info),
- h_t : total tooth/slot depth (**semi-open** : $h_{s0} + h_{s1} + h_{s2}$; **open slot** : h_s),
- d_{cs} : stator core (yoke) radial thickness (meters),
- z : depth coordinate measured from the bore.

For each segment $[z_0, z_1]$:

$$\phi_{mean} = \frac{\phi(z_0) + \phi(z_1)}{2} \tag{4.3}$$

$$B_{mean} = \frac{\phi_{mean}}{L_i\, w_{mean}} \tag{4.4}$$

$$AT_{seg} = H(B_{mean})\, h_{seg} \tag{4.5}$$

and then :

$$\boxed{ATST = \sum AT_{seg}} \tag{4.6}$$

The tab shows the intermediate values : $\phi(z_i)$, ϕ_{mean}, w_{mean}, B_{mean}, H, and each segment contribution.

4.6.1 Semi-open (trapezoidal) stator tooth segmentation (3 segments)

The program uses three regions :
— seg0 : depth h_{s0} with width w_{st0}
— seg1 : depth h_{s1} with width $w_{mean} = \frac{w_{st0}+w_{st}}{2}$
— seg2 : depth h_{s2} with width w_{st}

4.6.2 Open-slot stator tooth (1 segment)

For open slots only one segement is required a single segment across h_s with :

$$w_{mean} = \frac{w_{st1} + w_{st2}}{2}.$$

4.7 Step 4 : Rotor Core Ampere-Turns ATRC

Rotor mean diameter is computed in the tab as :

$$\boxed{D_{r,mean} = D_r - 2\,d_{rb} - W_{ry}} \tag{4.7}$$

(in mm in the rotor results, then converted to meters).

Rotor core ampere-turns :

$$\boxed{ATRC = \frac{\pi\, D_{r,mean}\, H}{P \times 3}} \tag{4.8}$$

4.8 Step 5 : Rotor Tooth Ampere-Turns ATRT

The rotor tooth uses an analogous depth-varying flux profile, but with rotor yoke depth w_{ry} :

$$\boxed{\phi(z) = \phi_{max}\left(1 - \frac{z}{h_t + 0.7\,w_{ry}}\right), \qquad h_t = h_{r0} + d_{rb}} \tag{4.9}$$

The rotor tooth is split into 3 segments exactly as the tab describes :
— seg0 : $0 \to h_{r0}$ with width = tooth width at bore w_{bore}
— seg1 : $h_{r0} \to h_{r0} + r_1$ with width mean = $\frac{w_{bore}+W_{rt}}{2}$
— seg2 : $h_{r0} + r_1 \to h_{r0} + d_{rb}$ with width = W_{rt} and depth $d_{rb} - r_1$
and it applies the same sequence :

$$\phi(z_i) \to \phi_{mean} \to B_{mean} = \frac{\phi_{mean}}{L_i w_{mean}} \to H(B_{mean}) \to AT_{seg}.$$

Finally :

$$\boxed{ATRT = AT_0 + AT_1 + AT_2} \tag{4.10}$$

4.9 Step 6 : Airgap Ampere-Turns ATg (Carter-corrected)

Computes Carter factors for stator and rotor :
— Stator : uses **bs** for open-slot or **bs0** for semi-open, with ratio (opening)/L_g and a corresponding Carter table.
— Rotor : uses **b_r0** and the semi-open Carter table.
The effective airgap length is :

$$\boxed{L_{gd} = L_g\, k_{gs}\, k_{gr}} \tag{4.11}$$

Airgap area per pole and airgap flux density :

$$A_{gp} = \frac{\pi D\, L_s(\text{corr})}{P} \tag{4.12}$$

$$B_g = \frac{\phi_{pole}(\text{corr})}{A_{gp}} \tag{4.13}$$

Then uses the exact numerical constant and factor shown :

$$\boxed{ATg = 0.796 \times 10^6\ B_g\ 1.36\ L_{gd}} \tag{4.14}$$

4.10 Step 7 : Total MMF AT_total

$$\boxed{AT_{total} = ATSC + ATST + ATRC + ATRT + ATg} \tag{4.15}$$

4.11 Step 8 : Magnetizing Current Im

Computes :

$$\boxed{I_m = \frac{P\, AT_{total}}{2 \times 1.17 \times k_w \times T_{ph}}} \tag{4.16}$$

where k_w is the winding factor used by the stator design module and T_{ph} is the turns per phase and P is the number of poles.

4.12 Step 9 : No-load losses and no-load current

4.12.1 Loss terms collected (as displayed)

The tab lists :
— Stator copper : P_{js} (from stator winding module)
— Stator iron : P_{it} (teeth), P_{ic} (core)
— Rotor copper : P_{bar} (bars), P_{er} (end rings)
— Friction/windage : $P_{fw} = 0.01 \times P_{elec} \times 1000$
— Rotor iron losses computed inside the tab : P_{itr} (rotor teeth) and P_{icr} (rotor core)

4.12.2 No-load power

$$\boxed{P_{nl} = P_{it} + P_{ic} + P_{fw} + P_{itr} + P_{icr}} \tag{4.17}$$

$$\boxed{P_{loss,total} = P_{nl} + P_{js} + P_{bar} + P_{er}} \tag{4.18}$$

4.12.3 No-load current and no-load power factor

Using the user line voltage V_{line} :

$$I_w = \frac{P_{nl}}{\sqrt{3}\,V_{line}} \tag{4.19}$$

$$I_0 = \sqrt{I_w^2 + I_m^2} \tag{4.20}$$

$$Pf_0 = \frac{I_w}{I_0} \tag{4.21}$$

4.13 Step 10 : Actual efficiency (from total losses)

Computes efficiency using the rated power entered by the user (P in kW) and total losses in kW :

$$\boxed{\eta = \frac{P}{P + P_{loss,total}}} \tag{4.22}$$

4.14 Step 11 : Nominal slip estimate g from losses

Estimates slip from losses using :

$$\boxed{g = \frac{P_{bar} + P_{er}}{P_{in} - P_{js} - (P_{it} + P_{ic})}} \tag{4.23}$$

where :

$$P_{in} = P_{elec} \times 1000$$

is taken from the stator electrical block (kW converted to W).

4.15 Summary Table (same variable names as the tab)

Table 4.6 – Main outputs

Quantity	**Displayed symbol/name**
Stator core MMF	ATSC
Stator tooth MMF	ATST
Rotor core MMF	ATRC
Rotor tooth MMF	ATRT
Airgap MMF	ATg
Total MMF	AT_total
Magnetizing current	Im
No-load power	Pnl
No-load current components	Iw, I0
No-load power factor	Pf0
Total losses	total_losses
Actual efficiency	efficiency
Nominal slip estimate	g

Chapitre 5

Application Presentation (2D and 3D Views)

This chapter presents an interactive design application for a squirrel-cage induction motor, developed using Python,HTML and JavaScript with the Canvas API for real-time visualization. The application enables the design and visualization of detailed cross-sectional views of the motor, including accurate stator and rotor slot geometries with multiple profile types.

5.1 Two-Dimensional (2D) Visualization

5.1.1 Geometrical Fundamentals of the Motor

Main Diameters

The basic dimensions of the motor are defined by four characteristic diameters :

$$D = \text{Stator bore diameter} \quad [\text{mm}] \tag{5.1}$$
$$D_{ext} = \text{External stator diameter} \quad [\text{mm}] \tag{5.2}$$
$$D_r = \text{Rotor outer diameter} \quad [\text{mm}] \tag{5.3}$$
$$D_{sh} = \text{Shaft diameter} \quad [\text{mm}] \tag{5.4}$$

The corresponding radii are given by :

$$R_{bore} = \frac{D}{2} \quad [\text{mm}] \tag{5.5}$$
$$R_{ext} = \frac{D_{ext}}{2} \quad [\text{mm}] \tag{5.6}$$
$$R_r = \frac{D_r}{2} \quad [\text{mm}] \tag{5.7}$$
$$R_{sh} = \frac{D_{sh}}{2} \quad [\text{mm}] \tag{5.8}$$

Magnetic Air Gap

The magnetic air gap between the stator and the rotor is defined as :

$$g = R_{bore} - R_r \quad [\text{mm}] \tag{5.9}$$

5.1.2 Stator Slot Geometry

Fundamental Parameters

The number of stator slots S_s is an integer that defines the angular discretization :

$$\theta_s = \frac{2\pi}{S_s} \quad [\text{rad}] = \frac{360°}{S_s} \quad [\text{degrees}] \tag{5.10}$$

The slot pitch at the stator bore is given by :

$$p_s = \frac{\pi D}{S_s} \quad [\text{mm}] \tag{5.11}$$

Type 1 : Rectangular Slots (Parallel-Sided Profile)

For parallel-sided (rectangular) slots, the defining parameters are :
— b_s : slot width [mm]
— h_s : slot depth [mm]
The angular position of the i-th slot is defined as :

$$\phi_i = \frac{2\pi i}{S_s} + \frac{\theta_s}{2} \quad \text{with } i = 0, 1, \ldots, S_s - 1 \tag{5.12}$$

The Cartesian coordinates of a slot point at radius r and angle ϕ are expressed as :

$$\begin{aligned} x &= r\cos(\phi \pm \Delta\phi) \\ y &= r\sin(\phi \pm \Delta\phi) \end{aligned} \tag{5.13}$$

where $\Delta\phi = \dfrac{b_s}{2r}$ represents the angular half-width of the slot.

Type 2 : Trapezoidal Slots

Trapezoidal slots are more complex and are composed of three distinct sections :

Section 1 : Slot Opening

$$b_{s0} : \text{opening width at the bore} \quad [\text{mm}] \tag{5.14}$$
$$h_{s0} : \text{opening depth} \quad [\text{mm}] \tag{5.15}$$

Section 2 : Slot Shoe

$$b_{s1} : \text{slot width at depth } h_{s0} + h_{s1} \quad [\text{mm}] \tag{5.16}$$
$$h_{s1} : \text{shoe height} \quad [\text{mm}] \tag{5.17}$$

Section 3 : Main Slot Body

$$b_{s2} : \text{slot width at the bottom} \quad [\text{mm}] \tag{5.18}$$
$$h_{s2} : \text{main body height} \quad [\text{mm}] \tag{5.19}$$

The total slot depth is expressed as :

$$h_s = h_{s0} + h_{s1} + h_{s2} \quad [\text{mm}] \tag{5.20}$$

For each trapezoidal section, the slot width is obtained by linear interpolation :

$$b(\Delta h) = b_{\text{start}} + \frac{b_{\text{end}} - b_{\text{start}}}{h_{\text{section}}} \cdot \Delta h \tag{5.21}$$

where Δh is the distance measured from the beginning of the section and h_{section} is the height of the considered section.

5.1.3 Rotor Slot Geometry

Fundamental Parameters

Similarly to the stator, the angular pitch of the rotor slots is defined as :

$$\theta_r = \frac{2\pi}{N_r} \quad [\text{rad}] \tag{5.22}$$

where N_r is the number of rotor slots.

Type 1 : Rectangular Rotor Slots

Rectangular rotor slots follow the same geometrical principles as those used for the stator :

$$b_r : \text{slot width} \quad [\text{mm}] \tag{5.23}$$

$$h_r : \text{slot depth} \quad [\text{mm}] \tag{5.24}$$

Type 2 : Teardrop Rotor Slots

Teardrop-shaped slots are characteristic of squirrel-cage induction motor rotors and are composed of two main sections :

Opening Section : Rounded Profile

$$h_0 : \text{opening depth} \quad [\text{mm}] \tag{5.25}$$

$$b_0 : \text{opening width} \quad [\text{mm}] \tag{5.26}$$

$$r_1 : \text{upper curvature radius} \quad [\text{mm}] \tag{5.27}$$

The rounded shape of the opening is modeled using a circular arc :

$$(x - x_c)^2 + (y - y_c)^2 = r_1^2 \tag{5.28}$$

where (x_c, y_c) denotes the center of curvature.

Main Body Section : Trapezoidal or Rectangular Profile

$$r_2 : \text{lower curvature radius} \quad [\text{mm}] \tag{5.29}$$

$$d_{rb} : \text{slot depth below the opening} \quad [\text{mm}] \tag{5.30}$$

The total rotor slot depth is given by :

$$h_r = h_0 + d_{rb} \quad [\text{mm}] \tag{5.31}$$

5.1.4 Tooth Height Calculation

The stator tooth height (stator yoke thickness) is expressed as :

$$h_{\text{tooth},s} = R_{ext} - R_{bore} - h_s \tag{5.32}$$

For the rotor (rotor yoke thickness), the corresponding expression is :

$$h_{\text{tooth},r} = R_r - R_{sh} - h_r \tag{5.33}$$

where h_r represents the total depth of the rotor slots.

5.1.5 Angular–Linear Conversions

To convert a linear dimension d into an angular width at radius r :

$$\theta_{\text{ang}} = \frac{d}{r} \quad [\text{rad}] \tag{5.34}$$

To convert an angular value θ into an arc length at radius r :

$$d_{\text{arc}} = r \cdot \theta \quad [\text{mm}] \tag{5.35}$$

These relations allow the transition from linear geometry (mm) to angular geometry (rad), which is essential for computing tooth pitches and slot widths.

5.1.6 Simplified Magnetic Characteristic Calculations

Iron Cross-Section

The iron cross-section at a stator tooth can be approximated as :

$$S_{\text{tooth}} \approx b_s \cdot h_s \quad [\text{mm}^2] \tag{5.36}$$

The total stator iron cross-section is given by :

$$S_{\text{iron},s} = S_{\text{bore}} \cdot (1 - k_{\text{slot}}) \tag{5.37}$$

where S_{bore} is the total bore area and k_{slot} is the slot fill factor (ratio of slot area to total area).

5.2 Three-Dimensional (3D) View

5.2.1 Equations and Fundamental Principles

5.2.2 Stator Geometry

Stator Diameters and Radii

In the code, the stator is defined by two main diameters :
— External diameter : D_{ext}
— Internal diameter : D_{int}
The corresponding radii are :

$$R_{\text{ext}} = \frac{D_{\text{ext}}}{2}, \qquad R_{\text{int}} = \frac{D_{\text{int}}}{2} \tag{5.38}$$

Stator Back Iron Thickness

The stator back iron corresponds to the iron region located between the bottom of the slots and the external diameter. Based on the input diameters and slot depth, the correct radial thickness is given by :

$$e_{\text{back iron}} = \frac{D_{\text{ext}} - D_{\text{int}}}{2} - h_{\text{slot}} \quad [\text{mm}] \tag{5.39}$$

where D_{ext} and D_{int} are the external and internal stator diameters, respectively, and h_{slot} is the stator slot depth.

Stator Slot Geometry

The number of stator slots N_s defines the angular discretization of the stator :

$$\theta_{\text{slot}} = \frac{2\pi}{N_s} \quad [\text{rad}] \tag{5.40}$$

The arc length corresponding to one slot, measured at the inner radius, is :

$$s_{\text{arc}} = \theta_{\text{slot}} \cdot R_{\text{int}} \tag{5.41}$$

The actual tooth width is obtained by subtracting the slot width w_{slot} from this arc pitch :

$$w_{\text{tooth}} = \max(0.5,\ s_{\text{arc}} - w_{\text{slot}}) \tag{5.42}$$

The $\max(0.5, \cdot)$ term corresponds to a safety constraint implemented in the code to prevent excessively thin teeth.
The slot depth is directly defined as an input parameter :

$$p_{\text{slot}} = h_{\text{slot}} \quad [\text{mm}] \tag{5.43}$$

where h_{slot} represents the radial depth of the stator slots.

5.2.3 Rotor Geometry

Rotor Diameter and Radius

In the code, the rotor is directly defined by its external diameter D_{rot}. The corresponding radius is :

$$R_{\text{rot}} = \frac{D_{\text{rot}}}{2} \quad [\text{mm}] \tag{5.44}$$

This radius must be smaller than the stator inner radius in order to create the air gap.

Rotor Back Iron Thickness

The rotor back iron is computed as the radial difference between the rotor radius and the rotor slot depth :

$$e_{\text{back iron},r} = R_{\text{rot}} - h_{\text{slot,rot}} \quad [\text{mm}] \tag{5.45}$$

where $h_{\text{slot,rot}}$ is the depth of the rotor slots.

5.2.4 Rotor Bar Geometry

Number and Distribution of Bars

The number of rotor bars N_b is defined as an input parameter. The angular spacing between bars is given by :

$$\theta_{\text{bar}} = \frac{2\pi}{N_b} \quad [\text{rad}] \tag{5.46}$$

Bar Dimensions and Positioning

Each rotor bar is modeled as a rectangular parallelepiped defined by :
— **Radial thickness** : t_{bar} (`bar_thickness` in the code)
— **Circumferential width** : w_{bar} (`bar_width`)
— **Axial length** : L_{bar} (`bar_length`)
The radial starting position of each bar is :

$$R_{\text{bar,start}} = R_{\text{rot}} - t_{\text{bar}} \tag{5.47}$$

The angular position of the i-th bar is :

$$\phi_i = i \cdot \theta_{\text{bar}} \quad i = 0, 1, \ldots, N_b - 1 \tag{5.48}$$

5.2.5 Air Gap

The air gap is computed directly from the stator internal diameter and the rotor external diameter :

$$g = \frac{D_{\text{int,stator}}}{2} - \frac{D_{\text{rot}}}{2} \quad [\text{mm}] \tag{5.49}$$

where :
— $D_{\text{int,stator}}$: internal stator diameter
— D_{rot} : external rotor diameter

5.2.6 Code Structure and Operation

5.2.7 Overall Architecture

The application follows a modular architecture composed of the following main components :

- **Main Class** : Manages the user interface and user interactions
- **Geometry Generator** : Computes 3D coordinates based on input parameters
- **Visualization Module** : Uses Plotly to display the interactive 3D model
- **Parameter Manager** : Validates and stores user inputs

5.2.8 Main Execution Workflow

Step 1 : Initialization and User Interface

The code creates an interactive web interface (HTML/JavaScript) that allows the user to enter motor parameters :

```
document.getElementById('stator_external_diameter').value = 256; // mm
document.getElementById('stator_num_slots').value = 36;
// integer

document.getElementById('rotor_bar_length').value = 332;
// mm
```

Step 2 : Parameter Validation and Preprocessing

All input parameters are checked for physical and geometrical consistency before proceeding with the geometry generation (valid ranges, non-negative values, and dimensional compatibility).

Step 3 : Stator Geometry Computation

The stator geometry is generated through the following steps :

1. Computation of internal and external radii
2. Calculation of the stator back iron thickness
3. Generation of rectangular stator slots
4. Construction of stator teeth (with width adjusted according to the angular pitch)

Step 4 : Rotor Geometry Computation

The rotor geometry is generated in a similar manner to the stator, including its own back iron and slots. The rotor bars are then positioned individually.

Step 5 : Rotor Bar Positioning

Each rotor bar i is positioned on a circle of radius R_{bars} according to :

$$\begin{aligned} x_i &= R_{\text{bars}} \cdot \cos\left(\frac{2\pi i}{N_b}\right) \\ y_i &= R_{\text{bars}} \cdot \sin\left(\frac{2\pi i}{N_b}\right) \end{aligned} \tag{5.50}$$

Step 6 : Assembly of the 3D Model

All geometrical components (stator, rotor, bars, and auxiliary structures) are combined into a single coherent 3D model.

Step 7 : Interactive Visualization

All elements (stator, rotor, bars, and structural components) are rendered within a Plotly.js 3D scene and displayed interactively. The user can :
— Rotate the model freely
— Zoom in and out
— Inspect the details of each component
— Export the model in various formats

5.2.9 Key Implementation Principles

Use of Polar Coordinates

The code systematically relies on polar coordinates to generate circular geometries (stator and rotor) :

$$\begin{aligned} x &= r\cos(\theta) \\ y &= r\sin(\theta) \\ z &= z \end{aligned} \tag{5.51}$$

This approach significantly simplifies the generation of repetitive circular structures.

Triangulated Surface Construction

The 3D surfaces are built from elementary triangular facets. For each surface, the code generates :

1. A set of 3D points (vertices)
2. A set of triangles connecting these points (faces)
3. Color and transparency attributes

5.2.10 Simplified Execution Flow

```
  Parameter Input
 (HTML User Interface)

                ↓

  Parameter Validation
  (Consistency Checks)

                ↓

 Stator Geometry Computation
(Slots, Teeth, Back Iron)
```

```
                ↓

  Rotor Geometry Computation
  (Back Iron + Rotor Bars)

                ↓

 Support Structure Generation
  (Bearings, Frame)

                ↓

  3D Model Assembly
  (Combination of All Elements)

                ↓

  Interactive Visualization
      (Plotly Rendering)
```

5.2.11 Geometrical Primitives Used

Rotated Radial Boxes (`mesh3d`)

Teeth and rotor bars are generated as rectangular parallelepipeds positioned at a given radius and rotated by an angle α :

$$x' = x\cos\alpha - y\sin\alpha, \quad y' = x\sin\alpha + y\cos\alpha, \quad z' = z$$

Solid and Hollow Cylinders (`surface`)

Yokes, rings, and cores are generated using parametric surfaces :

$$x(\theta) = R\cos\theta, \quad y(\theta) = R\sin\theta, \quad z \in \{z_0, z_0 + H\}$$

Both internal and external surfaces, as well as annular end disks, are used to close the volume.

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